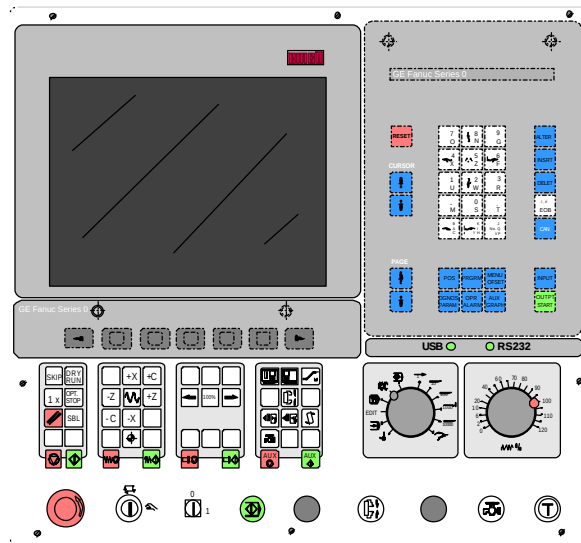


EMCO WinNC GE Series Fanuc 0-TC

Software description/ Software version from 13.76



Software description

EMCO WinNC Fanuc 0-TC

Ref.No. EN 1802 Edition H2003-7

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industrial training systems

Preface

The software EMCO WinNC FANUC SERIES 0-TC Turning is a part of the EMCO education concept on PC basis.

Target of this concept is learning to operate and program the original control at the PC.

The milling machines of the EMCO PC TURN und Concept TURN series can be directly controlled via PC by means of the EMCO WinNC for the EMCO TURN.

The operation is rendered very easy by the use of a digitizer or the control keyboard with TFT flat panel display (optional accessory), and it is didactically especially valuable since it remains very close to the original control.

This manual does not include the whole functionality of the control software FANUC SERIES 0-TC Turning, however emphasis was laid on the simple and clear illustration of the most important functions so as to achieve a most comprehensive learning success.

In case any questions or proposals for improving this manual should arise, please contact us directly:

EMCO MAIER Gesellschaft m. b. H.
Department Technical Documentation
A-5400 Hallein, Austria

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Starting Information

see attachment

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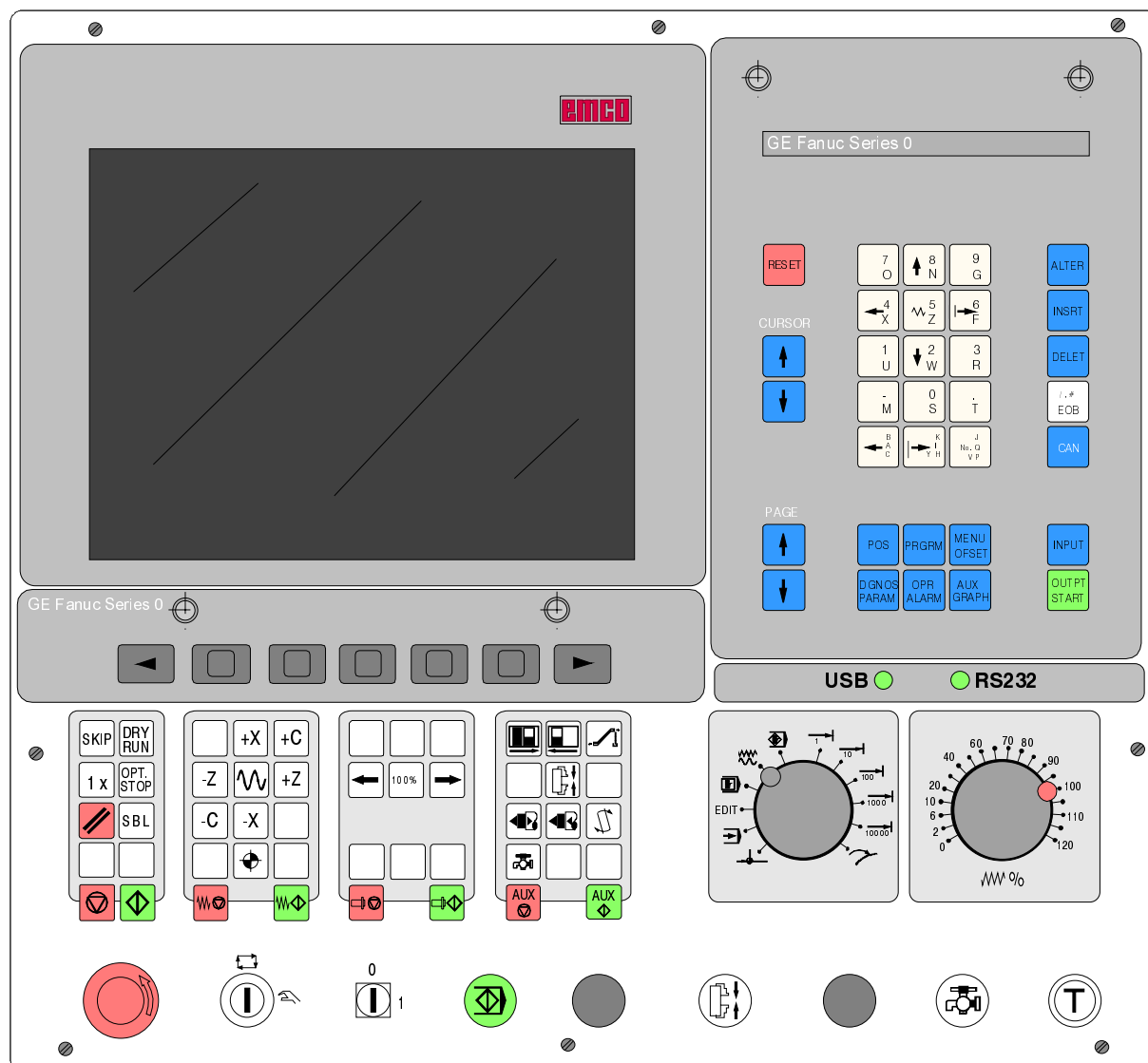
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A: Key Description

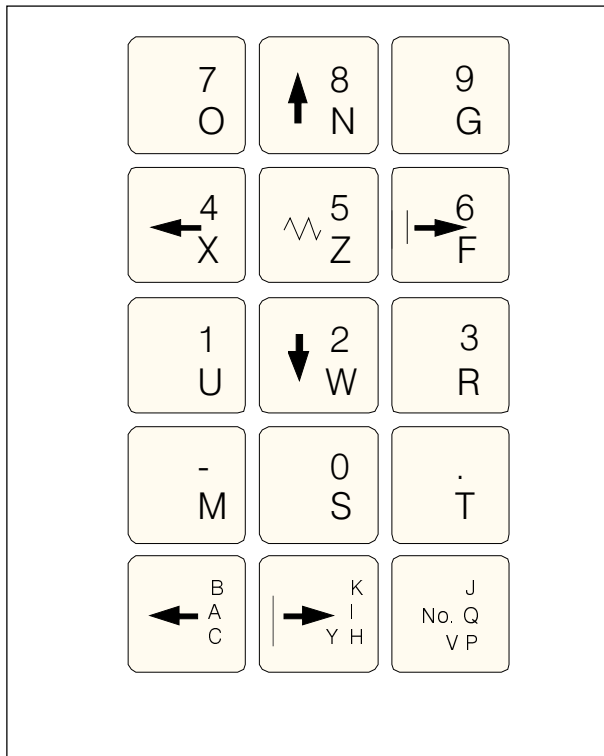
Control Keyboard, Digitizer Layout



Key Functions

RESET Cancel an alarm, reset the CNC
 (e.g. interrupt a program), etc.
 CURSOR Search function, line up/down,
 program call-up
 PAGE Page up/down
 ALTER Alter word (replace)
 INSERT Insert word, create new program
 DELETE Delete program, block, word
 /, # EOB Skip block, **End of Block**
 CAN Delete program, block, word
 INPUT Word input, data input
 OUTP START Start data output

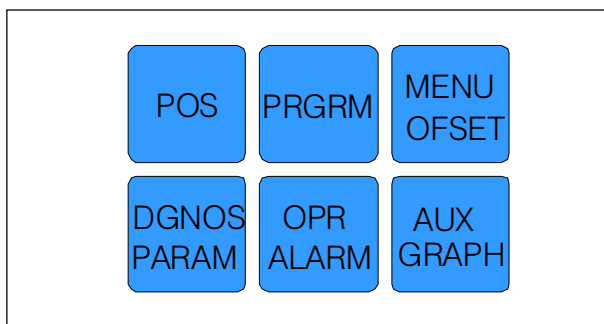
POS Indicates the current position
 PRGRM Edit and display of the program,
 input of the MDI data; display of
 command value in automatic
 mode
 MENU OFSET Setting and display of offset value,
 tool and wear data, variables
 DGNOS PARAM .. Setting and display of parameter
 and display of diagnostic data
 OPR ALARM Alarm and message display
 AUX GRAPH Graphic display



Data Input Keys

Note for data input keys

Each data input key runs several functions (number, address character(s)). Repeated pressing of the key switches to the next function automatically.



Function Keys

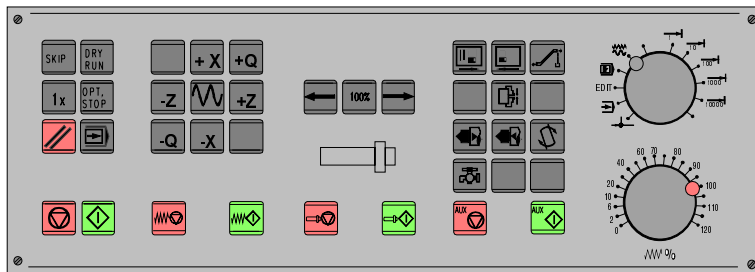
Note for function keys

At the PC keyboard the function keys can be displayed as softkeys with the key F12.

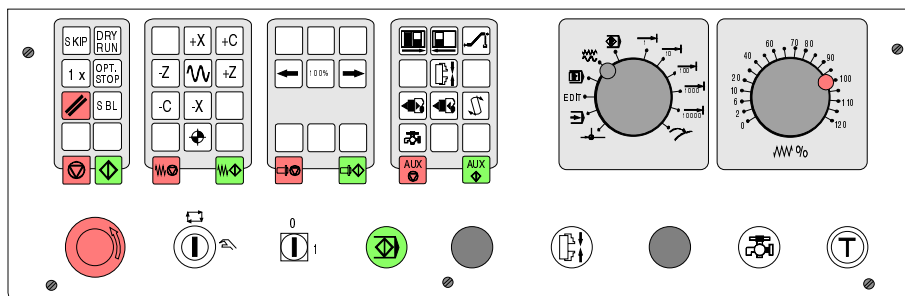
Machine Control Keys

The machine control keys are in the lower block of the control keyboard resp. the digitizer overlay.

Depending on the used machine and the used accessories not all functions may be active.

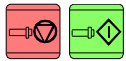


Machine control keyboard of the EMCO control keyboard



Machine control keyboard of the EMCO PC- Turn Series

	SKIP (skip blocks will not be executed)
	DRY RUN (test run of programs)
	OPT STOP (program stop at M01)
	RESET
	Single block machining
	Program stop / program start
	Manual axis movement
	Approaching the reference point in all axes
	Feed stop / feed start
	Spindle override lower / 100% / higher



Spindel stop / spindle start; spindle start in JOG and INC1...INC10000 mode:

Clockwise: press  key short, Counterclockwise: press  min. 1 sec.



Open / close door



Close / open clamping device



Tailstock back / forward



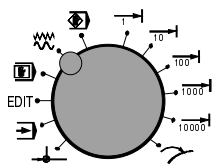
Swivel tool holder



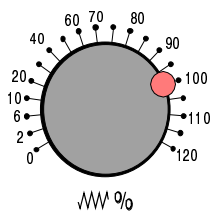
Coolant / puff blowing on / off



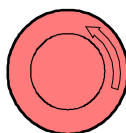
AUX OFF / AUX ON (auxiliary drives off / on)



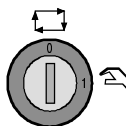
Mode selector



Feed / rapid feed override switch



EMERGENCY OFF (Unlock: pull out button)



Key switch for special operations (siehe Maschinenbeschreibung)



Additional NC start key



Additional key clamping device

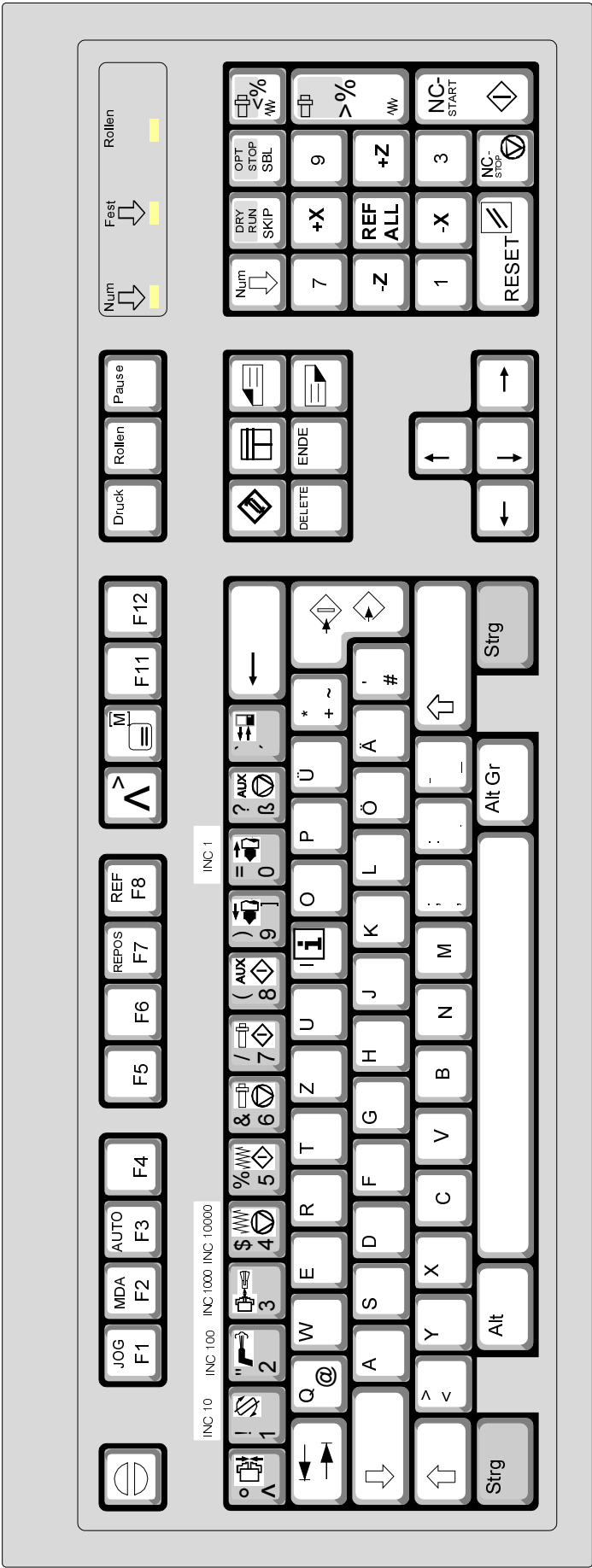


Consent key



No function

PC Keyboard



The machine functions in the numeric key block are active only with active NUM lock.

* With F12 the function keys POS, PROG, OFFSET SETTING, SYSTEM, MESSAGES and GRAPH will be displayed in the softkey line.

Some alarms will be acknowledged with the key ESC. By pressing the key F1 the modes (MEM, EDIT, MDI,...) will be displayed in the softkey line.

The assignment of the accessory functions is described in the chapter "Accessory Functions".

The meaning of the key combination ctrl 2 depends on the machine:
EMCO PC TURN 50/55: Puff blowing ON/OFF
EMCO PC TURN 100/125/155: coolant ON/OFF



B: BASICS

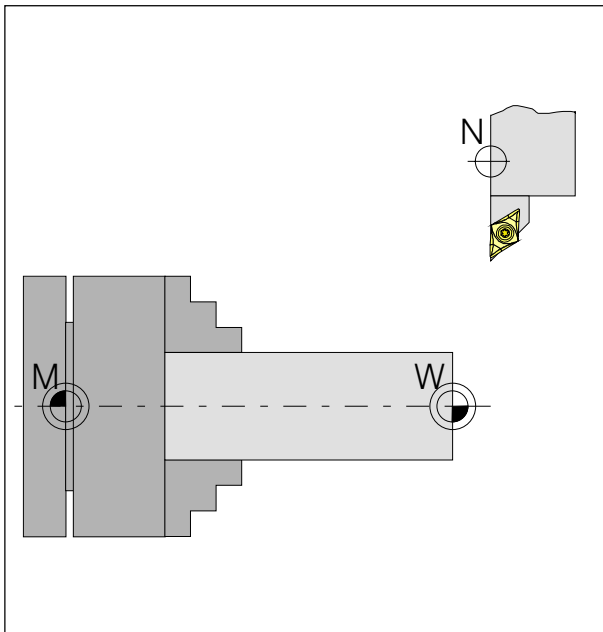
Reference Points of the EMCO Lathes

M = Machine zero point

An unchangeable reference point established by the machine manufacturer.

Proceeding from this point the entire machine is measured.

At the same time "M" is the origin of the coordinate system.



Reference points in the working area

R = Reference point

A position in the machine working area which is determined exactly by limit switches. The slide positions are reported to the control by the slides approaching the "R".

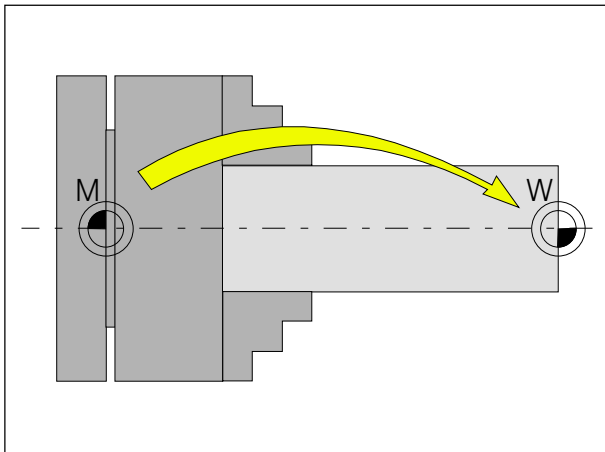
Required also after every power failure.

N = Tool mount reference point

Starting point for the measurement of the tools. "N" lies at a suitable point on the tool holder system and is established by the machine manufacturer.

W = Workpiece zero point

Starting point for the dimensions in the part program. Can be freely established by the programmer and moved as desired within the part program.



Zero offset from machine zero point M to workpiece zero point W

Zero Offset

With EMCO lathes the machine zero "M" lies on the rotating axis and on the end face of the spindle flange. This position is unsuitable as a starting point for dimensioning. With the so-called zero offset the coordinate system can be moved to a suitable point in the working area of the machine.

The offset register offers one adjustable zero offset.

When you define a value in the offset register, this value will be considered with program start and the coordinate zero point will be shifted from the machine zero M to the workpiece zero W.

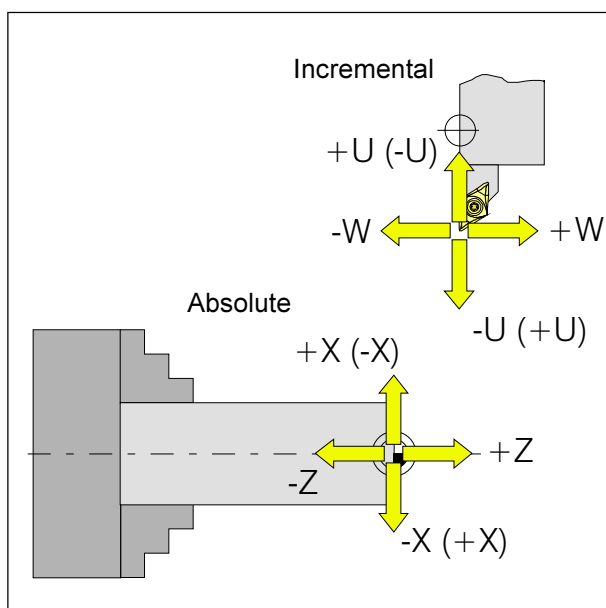
The workpiece zero point can be shifted within a program with "**G92 - Coordinate system setting**" in any number.

More informations see in the command description.

The Coordinate System

The X coordinate lies in the directions of the cross slide, the Z coordinate in the direction of the longitudinal slide.

Coordinate values in minus directions describe movements of the tool system towards the workpiece. Values in plus direction away from the workpiece,



Absolute coordinates refer to a fixed position, incremental coordinates to the tool position. The bracket values for X, -X, U, -U are valid for the PC TURN 50/55 because the tool is in front of the turning centre on this machine.

Coordinate System for Absolute Value Programming

The origin of the coordinate system lies at the machine zero "M" or at the workpiece zero "W" following a programmed zero offset.

All target points are described from the origin of the coordinate system by the indication of the respective X and Z distances.

X distances are indicated as the diameter (as dimensioned on the drawing).

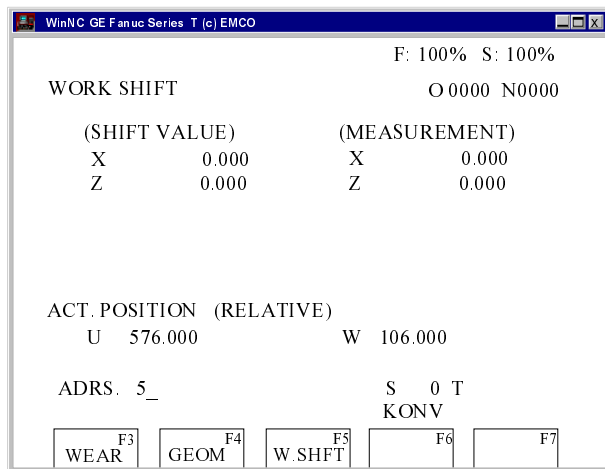
Coordinate System for Incremental Value Programming

The origin of the coordinate system lies at the tool mount reference point "N" or at the cutting tip after a tool call-up.

The U coordinate lies in the direction of the cross slide, the W coordinate in the direction of the longitudinal slide. The plus and minus directions are the same as for absolute value programming.

With incremental value programming the actual paths of the tool (from point to point) are described.

X distances are indicated as the diameter.



Input pattern for the zero offset

Input of the Zero Offset

- Press the key 
- Select the softkey W. SHFT (work shift)
- The input pattern beside appears
- Below (SHIFT VALUE) X, Z you can enter the offset **from the workpiece zero point to the machine zero point (neg. sign).**
- Enter the offset (e.g.: Z-30.5) and press the key

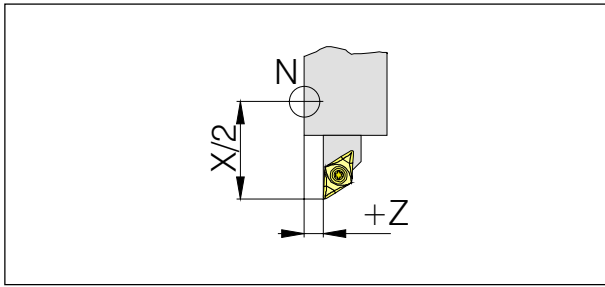


- This offset is always active (without separate call-up).

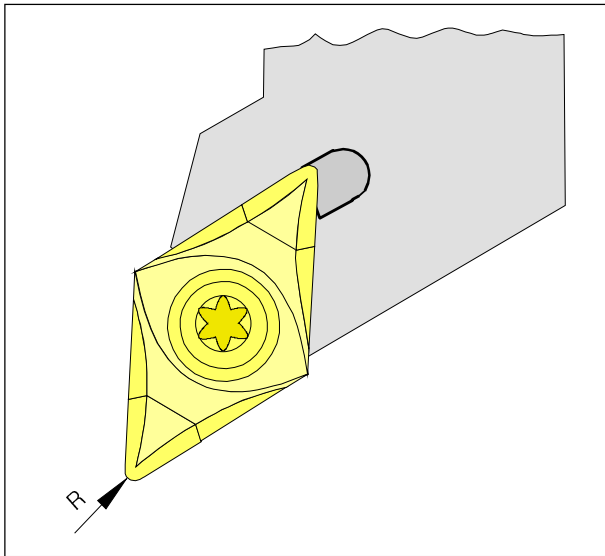
Note:

With this offset normally the coordinate zero will be shifted from the spindle flange to the stop face of the clamping device.

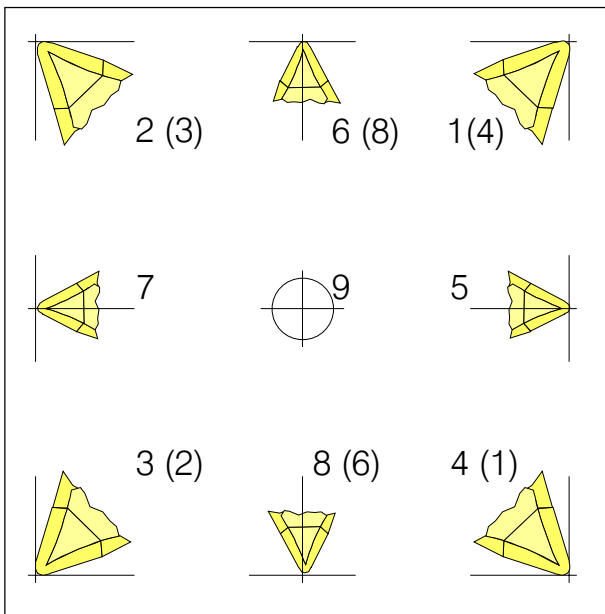
The work piece length (zero shift to the right work piece face) will be considered in the program with G92.



Length correction



Radius of the cutter tip R



Cutter position T

Look at the tool like it is clamped at the machine to determine the cutter position. For machines on which the tool is below (in front of) the turning centre (e.g. PC TURN 50/55) use the values in brackets because of the opposite +X direction.

Tool Data Measuring

Aim of the tool data measuring:

The CNC should use the tool tip for positioning, not the tool mount reference point.

Every tool which is used for machining has to be measured. The distances in both axis directions between tool tip and tool mount reference point "N" are to be measured.

In the so-called tool register the measured length corrections, the cutter radius and the cutter position can be stored.

(standard = 16)

The correction number can be any register number, but has to be considered with tool call in program.

Example

The length corrections of a tool in the tool turret station 4 have been stored as correction number 4.

Tool call in program: **T0404**

The first two numbers of the T word mark the position in the tool turret, the two last numbers mark the correction number belonging to it.

The length corrections can be measured half-automatically, **cutter radius** and **cutter position** have to be inserted manually.

Inserting cutter radius and cutter position is only necessary for using cutter radius compensation with this tool.

Tool data measuring occurs for

X in diameter

Z absolute from point "N"

R radius of the cutter tip

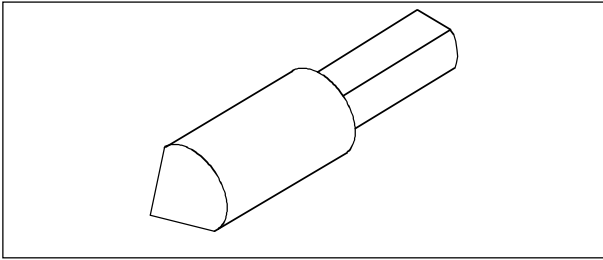
T cutter position

With "**offset wear**" occurs the correction of not exact measured tool data or of worn tools after several machining runs. The inserted length corrections will be added to or subtracted from the geometry of the tool incrementally.

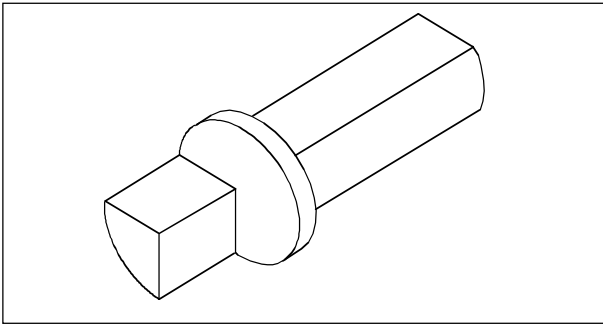
X+/- incremental in diameter to the value of the geometry

Z+/- incremental to the value of the geometry

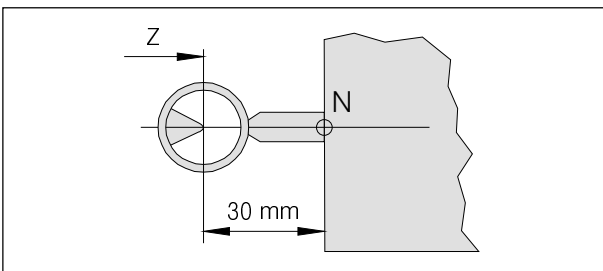
R+/- incremental to the value of the geometry



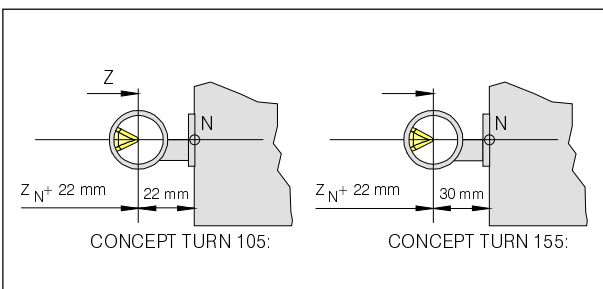
Reference tool Concept TURN 50/55



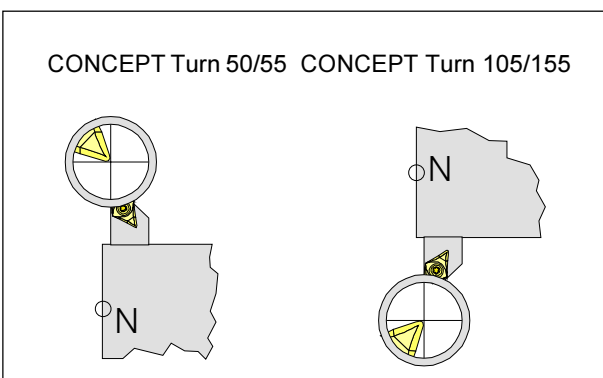
Reference tool Concept TURN 105/155



Reference tool measuring Concept Turn 50/55



Reference tool measuring Concept Turn 105/155



Traverse into the graticule with the tool

Tool Data Measuring with the Optical Presetting Device

Procedure

- Set up the optical presetting device.
The display in the optics is side-inverted.
- Clamp the reference tool into the tool turret disc (onto stop).
- In the JOG mode traverse the reference tool into the graticule of the optical presetting device.
- Press the **POS** key and select the softkey REL.
- Press key sequence **1 U** and **CAN** (U value is cancelled).
- Press key sequence **2 W** and **CAN** (W value is cancelled).
- Select JOG mode, reduce feed and traverse in W to -30 for the Concept Turn 50/55/155 or -22 for the Concept Turn 105 (20 mm tool length + 2 mm holder) (outstanding length of the reference tool).
- Reset W value to 0 (**2 W** and **CAN**).
Now the face of the tool holder disk is set to 0.
- Swivel in the tool and traverse it into the graticule.
- Press the **MENU OFFSET** key.
- Select the softkey GEOM.
- Select tool station number with CURSOR keys **↑** **↓**.

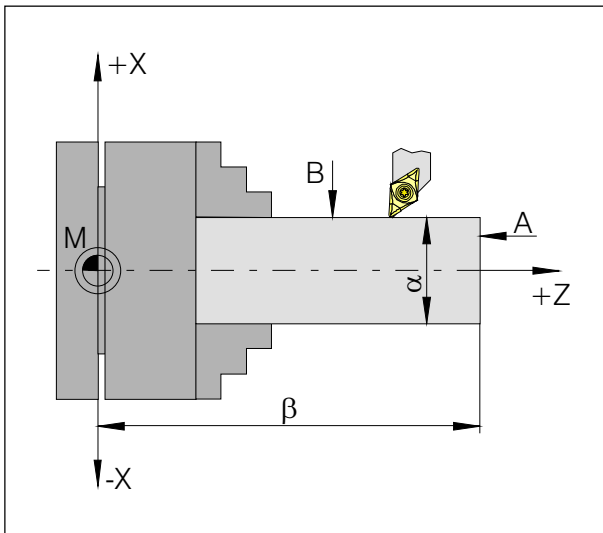
For X correction

- Press the key sequence **4 X** and **INPUT**.
- U value (difference between tool tip and point N) is transferred into tool data memory.

For Z correction

- Press the key sequence **5 Z** and **INPUT**.
- W value (difference between tool tip and point N) is transferred into tool data memory.
- Repeat this procedure for the following tools.

Tool Data Measuring by Scraping



Measures for scraping:

- A scraping on face
- B scraping on circumference
- α workpiece diameter
- β workpiece length + chuck length

- Measure the distance β (work piece length + chuck) and the diameter α of the workpiece.
- Start the spindle in MDI-Mode (M03/M04 S).
If you want to scrap with standing spindle you have to put a piece of paper between tool and workpiece. Take over the value when the tool is so near that you just can not move the paper any more.
- Select the mode JOG.
- Travel with the face of the tool holder disk (feed 1%) using the direction keys to the face A of the workpiece.
- Press the **POS** key and the softkey REL.
- Press key sequence **↓ 2 W** and **CAN** (W value is cancelled).
- Swivel the tool to be measured in working position.
- Travel with the tool tip using the direction keys to the face A of the workpiece (scraping)
- Press the **MENU OFFSET** key.
- Select the softkey GEOM.
- Select tool station number with CURSOR keys **↑** **↓**.
- Press the key sequence **↖ 5 Z** and **INPUT**.
- W value (difference between tool tip and point N) is transferred into tool data memory.
- Travel with the tool tip using the direction keys to the circumference B of the workpiece (scraping)
- Press the **POS** key and the softkey ABS.
- Subtract the diameter α of the work piece from the displayed value.
- Press the **MENU OFFSET** key.
- Select the softkey GEOM.
- Enter: **← 4 X**, the result of the subtraction and press **INPUT**.
- Repeat this sequence for the following tools.

C: Operating Sequences

Survey Operating Modes

ZRN

In this operating mode the reference point will be approached.

With reaching the reference point the actual position display is set to the value of the reference point coordinates.

By that the control acknowledges the position of the slides in the working area.

With the following situations the reference point has to be approached:

- After switching on the machine
- After mains interruption
- After alarm "Approach reference point" or "Ref. point not reached"
- After collisions or if the slides stuck because of overload

AUTO

For working off a part program the control calls up block after block and interprets them.

The interpretation considers all correction which are called up by the program.

The so-handled blocks will be worked off one by one.

EDIT

In the EDIT mode you can enter part programs and transmit data.

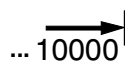
MDI

In the MDI mode you can switch on the spindle and swivel the tool holder.

The control works off the entered block and deletes the intermediate store for new inputs..

JOG

With the JOG keys the slides can be traversed manually.

S1 ... S1000  ... 

In this operation mode the slides can be traversed for the desired increment (1...1000 in $\mu\text{m}/10^{-4}$ inch) by

means of the JOG keys    .

The selected increment (1, 10, 100, ...) must be larger than the machine resolution (lowest possible traverse movement), otherwise no movement occurs.

REPOS

Repositioning, approach back to the contour in JOG mode.

Teach In

Making programs in dialogue with the machine in MDA mode.

Approach the Reference point

By approaching the reference point the control will be synchronized to the machine.

- Change into ZRN mode.
- Actuate first the direction keys  or , then  or  to approach the reference point in the respective direction.
- With the  key both axes will be approached automatically (PC keyboard).

Danger of collisions

Mind for obstacles in the working area (clamping devices, clamped work pieces, etc.).

After reaching the reference point its position will be displayed as actual position. Now the machine is synchronized to the control.

Input of the Gear Position

(only with EMCO PC Turn 50/55)

For that the machine runs the correct spindle speed, the selected gear (belt) position of the machine has to be entered in EMCO WinNC.

- Press the key .
- Press the key  multiple, until the setting page (GENERAL) will be displayed.
- Move the cursor on the input field "GEAR" and enter the corresponding gear position.

1 gear position 1	120 - 2000 U/min
2 gear position 2	280 - 4000 U/min

Setting of Language and Workpiece Directory

- Press the key .
- Press the key  multiple, until the setting page (GENERAL) will be displayed.

Workpiece Directory

In the workpiece directory the CNC programs created by the operator will be stored.

The workpiece directory is a subdirectory of the directory which was determined with installation.

Enter in the input field "PATH = ..." the name of the workpiece directory with the PC keyboard, max. 8 characters, no drives or pathes. Not existing directories will be created.

Active Language

Selection from installed languages, the selected language will be activated with restart of the software. Enter the language sign in the input field "LANG. = ...".

- DT for German
- EN for English
- FR for French
- SP for Spanish

Input of Programs

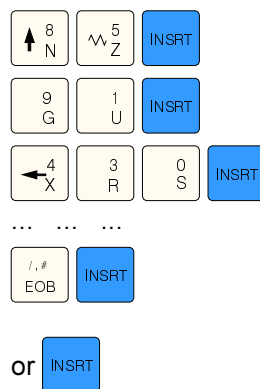
Part programs and subroutines can be entered in the EDIT mode..

Call up a program

- Change to the EDIT mode
- Press the  key
- With the softkey LIB the existing programs will be displayed
- Enter program number O...
The program numbers higher than 9500 are reserved for internals and must not be used
- New Program: Press the  key
- Existing program: Press the  key

Input of a block

Example:

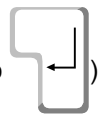


Block number (not necessary)

1. Word

2. Word

EOB - End of Block (on PC keyboard also



Note:

With the parameter "SEQ" (SETTING 1) an automatic block numbering can be set (1 = yes, 0 = no).

Search word

Enter the address of the word to be searched (e.g.: X)

and press the  key.

Insert word

Move the cursor before the word, that should be before the inserted word, enter the word (address and value) and press the  key.

Change word

Move the cursor before the word, that should be changed, enter the word and press the  key.

Delete word

Move the cursor before the word, that should be deleted and press the  key. Insert block

Move the cursor before the EOB sign ";" in that block, that should be before the inserted block and enter the block to be inserted.

Delete block

Enter block number (if no block number exists: N0) and press the  key.

Delete Program

EDIT mode

Enter the program number (e.g. O22) and press the

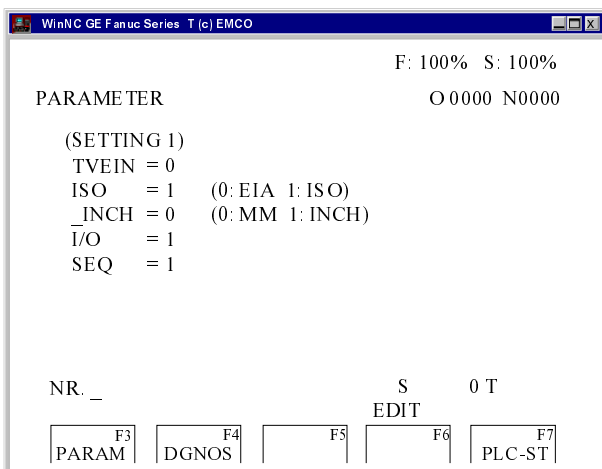
key .

Delete all Programs

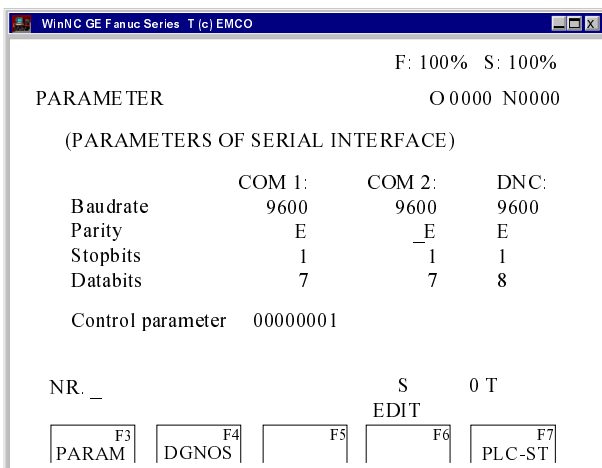
EDIT mode

Enter the program number 0 - 9999 and press the key

.



Selection of the input/output interface



Adjusting the serial interface

NOTE

When you use an interface expansion card (e.g. for COM 3 and COM 4), take care that for every interface a separate interrupt is used (e.g.: COM1 - IRQ4, COM2 - IRQ3, COM3 - IRQ11, COM4 - IRQ10).

Data Input - Output

- Press the key .

The screen shows (SETTING 3).

- Below "I/O" you can enter a serial interface (1 or 2) or a drive (A, B or C).
 - 1 serial interface COM1
 - 2 serial interface COM2
 - A disk drive A
 - B disk drive B
 - C hard disk drive C, workpiece directory (Established with installation or in (GENERAL)).
 - P Printer on LPT1.

Adjusting the Serial Interface

- Press the key .

- Press the key , until (PARAMETERS OF SERIAL INTERFACE) is displayed.

Settings:

Baudrate 110, 150, 300, 600, 1200, 2400, 4800, 9600

Parity E, O, N

Stopbits 1, 2

Datenbits 7, 8

Data transmission from / to original control in ISO-Code only

ISO: 7 Datenbits, Parity even (=E)

Control parameter:

Bit 0: 1...Transmission will be cancelled with ETX (End of Text) code

0...Transmission will be cancelled with RESET

Bit 7: 1...Overwrite part program without message

0...Message, if a program already exists

ETX code: % (25H)

Program Output

- EDIT mode
- Enter the receiver in (SETTING 1) below "I/O".
- Press the key .
- Enter the program to be sent (e.g. O22)
- When you enter e.g. O5-15, all programs with the numbers 5 to inclusive 15 will be transmitted. When you enter O-9999 as program number, all programs will be transmitted.
- Press the key .

Program Input

- EDIT mode
- Enter the receiver in (SETTING 1) below "I/O".
- Press the key .
- With input from disk or hard disk you have to enter a program number.
Enter the program number when you want to read in one program (e.g.: O22).
When you enter e.g. O5-15, all programs with the numbers 5 to inclusive 15 will be transmitted.
When you enter O-9999 as program number, all programs will be transmitted.
- Press the key .

Tool Offset Output

- EDIT mode
- Enter the receiver in (SETTING 1) below "I/O".
- Press the keys  and .
- If a drive (A, B or C) is the receiver, the zero offsets will be transmitted additionally.

Tool Offset Input

- EDIT mode
- Enter the sender in (SETTING 1) below "I/O".
- Press the keys  and .

Print Programs

- The printer has to be connected to LPT1 and must be in ON LINE status.
- EDIT mode
- Enter P (printer) in (SETTING 1) below "I/O".
- Press the key .
- Enter the program to be print (e.g. O22)
- When you enter e.g. O5-15, all programs with the numbers 5 to inclusive 15 will be printed. When you enter O-9999 as program number, all programs will be printed.
- Press the key .

Program Run

Start of a Part Program

Before starting a program the control and the machine must be ready for running the program.

- Select the EDIT mode.
- Press the key .
- Enter the desired part program number.
- Press the key .
- Change to AUTOMATIC mode.
- Press the key .

Displays while Program Run

While program run different values can be shown.

- Press the softkey PRGRM (basic status). While program run the actual program block will be displayed.
- Press the softkey CHECK. While program run the actual program block, the actual positions, active G and M commands and speed, feed and tool will be displayed.
- Press the key . The positions will be shown enlarged at the screen.

Block Search

With this function you can start a program at any block.

While block search the same calculations will be proceeded as with normal program run but the slides do not move.

- EDIT mode
- Select the program to be machined.
- Move the cursor with the keys  and  on that block, with which machining should start.
- Change to AUTOMATIC mode.
- Start the program with the key .

Program Influence

DRY RUN:

DRY RUN is used for testing programs. The main spindle will not be switched on and all movements occur in rapid feed.

If DRY RUN is active, DRY will be displayed in the first line on the screen.

SKIP:

With SKIP all program blocks which are marked with a "/" (e.g.: /N0120 G00 X...) will not be proceeded and the program will be continued with the next block without a "/" sign.

If SKIP is active, SKP will be displayed in the first line on the screen.

Program Interruption

Single block mode

After every program block the program will be stopped.

Continue the program with the key .

M00:

After M00 (programmed stop) in the program the program will be stopped. Continue the program with

the key .

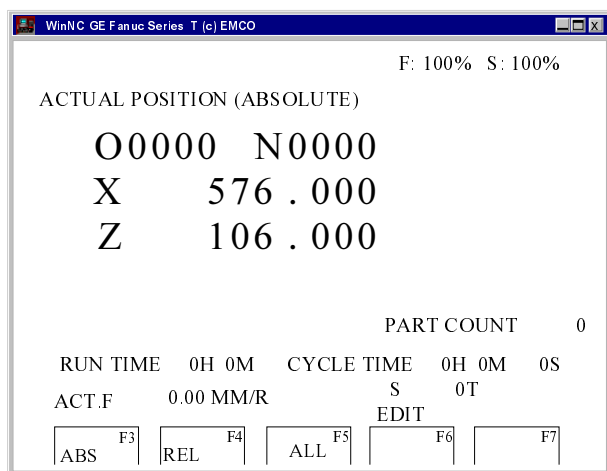
M01:

If OPT. STOP is active, (display OPT in the first line of the screen) M01 works like M00, otherwise M01 has no effect.

Display of the Software Versions

- Press the key .
- Select softkey DGNOS

The software version of EMCO WinNC and the eventually connected RS485 devices will be displayed.



Display of part counter and piece time

Part Counter and Piece Time

Below the position display the part counter and the piece time are displayed.

The part counter shows the number of program runs. Each M30 (or M02) increases the part counter for 1.

RUN TIME shows the complete running time of all program runs.

CYCLE TIME shows the running time of the actual program and will be reset to 0 with every program start.

Part Counter Reset

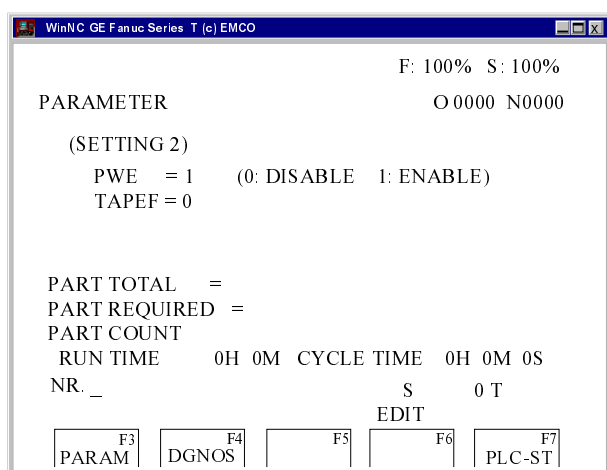
The part counter will be reset to 0 by pressing



(P) and  after.

RUN TIME Reset

The RUN TIME (total time) will be reset to 0 by pressing  (R) and  after.



Presetting the piece counter

Preset of the Part Counter

The part counter can be preset in (SETTING 2). Therefore move the cursor on the desired value and enter the new value.

PART TOTAL:

Each M30 increases this number by 1. Every program run of every program will be counted (= number of all program runs).

PART REQUIRED:

Preset part number. When this number is reached the program will be stopped and message 7043 PIECE COUNT REACHED will be displayed.

After that the program can be started only after resetting the part counter or increasing the preset part number.

Graphic Simulation

NC-programs can be simulated graphically.

Press the key .

The screen shows the input pattern for graphic simulation illustrated beside.

The simulation area is a rectangular window, which is determined by the right upper and left lower edge.

Inputs:

WORK LENGTH W
WORK DIAMETER D

Enter here the right upper edge of the simulation area.

GRAPHIC MINIMUM X, Z

Enter here the left lower edge of the simulation area.

All other inputs and the softkeys ZOOM and AUX are not active.

By pressing the key  the softkey 3DVIEW will be displayed.

Win 3D View is an option and not included in the basic version of the software.

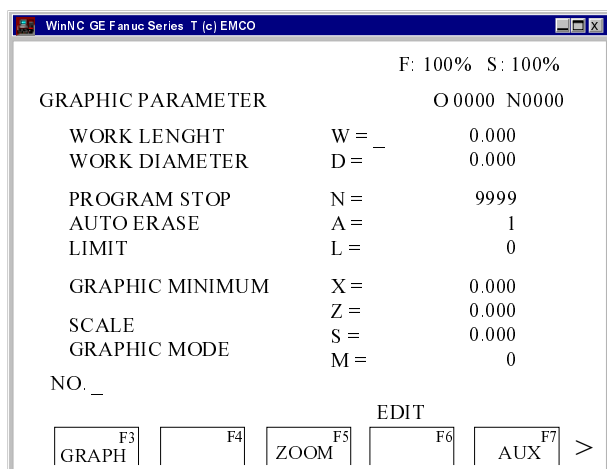
Press the softkey GRAPH to get into the simulation window.

The softkeys ZOOM, NORMAL and AUX are not active.

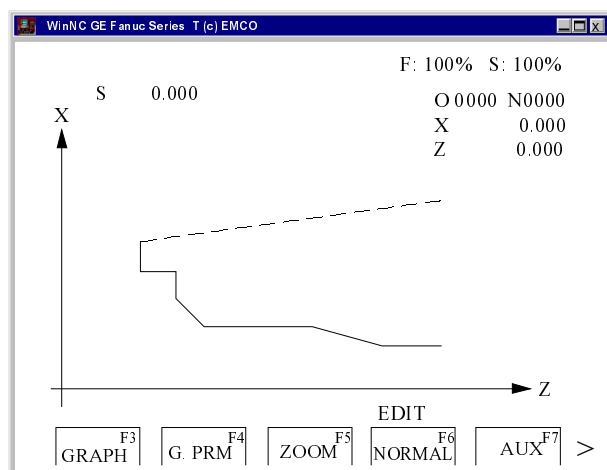
With the softkey G. PRM you will go back to the input pattern for graphic simulation.

With the key START the graphic simulation starts.
With the key STOP the graphic simulation stops.
With the key RESET the graphic simulation will be aborted.

Movements in rapid traverse will be displayed as dashed lines, movements in working traverse will be displayed as full lines.



Input pattern for graphic simulation



Simulation window

D: Programming

Program Structure

CNC programming for machine tools according to DIN 66025 is used.

The CNC program is a sequence of program blocks which are stored in the control.

With machining of workpieces these blocks will be read and checked by the computer in the programmed order.

The corresponding control signals will be sent to the machine.

The CNC program consists of:

- program number
- CNC blocks
- words
- addresses
- number combinations (for axis addresses partly with sign)

Used Addresses

O program number 1 to 9499 for part programs and subroutines

N block number 1 to 9999

G path function

X, Z position data in absolute values (X also dwell time)

U, W .. position data in incremental values (U also dwell time)

R radius, taper value, cycle parameter

C chamfer

I, K circle parameter

F feed rate, thread pitch

S spindle speed, cutting speed

T tool call (tool correction)

M miscellaneous function

P dwell, subprogram call, cycle parameter

Q cycle parameter

; block end

Survey of G Commands for Command Definition A, B, C

With software installation you can select the command definition A, B or C.

The difference between the versions is only the code for a command, but not the function of the command (see table).

In this manual only the command definition C is described.

If you use the command definition A or B, note the codes in the command description.

Gr.	Command			Function
	A	B	C	
0	+	G04		Dwell
	+	G07.1		Cylindrical Interpolaton
		G10		Data setting
		G11		Data setting Off
	+	G28		Return to reference point
	+	G70	G72	Finishing cycle
	+	G71	G73	Stock removal in turning
	+	G72	G74	Stock removal in facing
	+	G73	G75	Pattern repeating
	+	G74	G76	Deep hole drilling, cut-in cycle in Z
	+	G75	G77	Cut-in cycle in X
	+	G76	G78	Multiple threading cycle
1		G50	G92	Coord.syst. set., spindle speed limit
	•	G00		Positioning (rapid traverse)
		G01		Linear interpolation
		G02		Circular interpolation clockwise
		G03		Circular interp. counterclockwise
		G90	G77	G20 Longitudinal turning cycle
		G92	G78	G21 Thread cutting cycle
		G94	G79	G24 Face turning cycle
		G32	G33	Thread cutting
		G96		Constant cutting speed
2	•	G97		Direct spindle speed programming
3	•	-	G90	Absolute programming
	•	-	G91	Incremental programming
5		G98	G94	Feed per minute
	•	G99	G95	Feed per revolution
6		G20	G70	Inch data input
		G21	G71	Metric data input
7	•	G40		Cancel cutter radius compensation
		G41		Cutter radius compensation left
		G42		Cutter radius compensation right
10	•	G80		Cancel cycles
		G83		Drilling cycle
		G84		Tapping cycle
		G85		Reaming cycle
11	•	-	G98	Return to initial plane
	-	-	G99	Return to withdrawal plane
16		G17		Plane selection XY
		G18		Plane selection ZX
		G19		Plane selection YZ
21		G12.1		Polar Coordinate Interpolation ON
		G13.1		Polar Coordinate Interpolation OFF

- Initial status
- + Blockwise effective

With version A the commands of group 3 and 11 does not exist. Incremental programming occurs with version A always with U and W, Retraction movements occur always to the initial plane.

Survey of G Commands for Command Definition C

G00		Positioning (rapid traverse)
G01		Linear interpolation (feed)
G02		Circular interpolation clockwise
G03		Circular interpolation counterclockwise
G04+		Dwell
G7.1		Cylindrical Interpolaton
G10		Data setting
G11		Data setting Off
G12.1		Polar Coordinate Interpolation ON
G13.1		Polar Coordinate Interpolation OFF
G17		Plane selection XY
G18		Plane selection ZX
G19		Plane selection YZ
G20		Longitudinal turning cycle
G21		Thread cutting cycle
G24		Face turning cycle
G28+		Return to reference point
G33		Thread cutting
G40•		Cancel cutter radius compensation
G41		Cutter radius compensation left
G42		Cutter radius compensation right
G70		Inch data input
G71		Metric data input
G72+		Finishing cycle
G73+		Stock removal in turning
G74+		Stock removal in facing
G75+		Pattern repeating
G76+		Deep hole drilling, cut-in cycle in Z
G77+		Cut-in cycle in X
G78+		Multiple threading cycle
G80•		Cancel cycles (G83 up to G85)
G83		Drilling cycle
G84		Tapping cycle
G85		Reaming cycle
G90•		Absolute programming
G91		Incremental programming
G92		Coordinate system setting, spindle speed limit
G94		Feed per minute
G95•		Feed per revolution
G96		Constant cutting speed
G97•		Direct spindle speed programming
G98•		Return to initial plane
G99		Return to withdrawal plane

- Initial status
- + Blockwise effective

M- Commands

COMMAND	MEANING
M0	Programmed stop
M1	Programmed stop, conditional
M2	Program end
M3	Spindle ON clockwise
M4	Spindle ON counterclockwise
M5	Spindle OFF
M8	Coolant ON
M9	Coolant OFF
M13	Driven tools ON clockwise
M14	Driven tools ON counterclockwise
M15	DRIVEN TOOLS OFF
M20	Quill FORWARD
M21	Quill BACKWARD
M23	Collecting tray BACKWARD
M24	Collecting tray FORWARD
M25	Clamping device OPEN
M26	Clamping device CLOSE
M30	Main program end
M32	Program end for loading operation
M52	Round axis operation (C-axis ON)
M53	Spindle operation (C-axis OFF)
M57	Spindle oscillate ON
M58	Spindle oscillate OFF
M67	Bar feed / loading magazine feed ON
M68	Bar feed / loading magazine feed OFF
M69	Bar change
M71	Blow-out ON
M72	Blow-out OFF
M90	Manual clamping chuck
M91	Tensile clamping device
M92	Pressure clamping device
M93	Final position monitoring OFF
M94	Bar feed / loading magazine activate
M95	Bar feed / loading magazine deactivate
M98	Subroutine call
M99	Subroutine end, jump command

Description of G Commands

G00 Positioning (Rapid Traverse)

Format

N.... G00 X(U)... Z(W)...

The slides are traversed at maximum speed to the programmed target point. Incremental and absolute commands can be used at the same time.

Note

- A programmed feed F will be suppressed while G00.
- The maximum feed is defined by the producer of the machine.
- The feed override switch is limited to 100%.

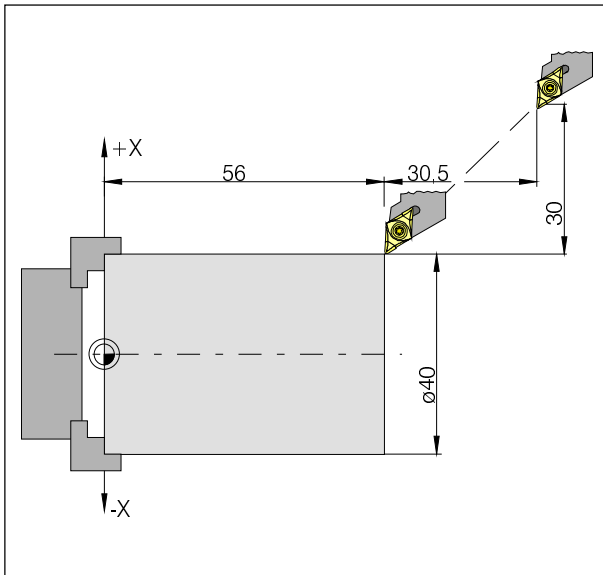
Example

absolute G90

N50 G00 X40 Z56

incremental G91

N50 G00 U-30 W-30.5



Absolute and incremental measures for G00

G01 Linear Interpolation (Feed)

Format

N... G01 X(U)... Z(W)... F....

Linear slide movements (face, longitudinal, taper turning) at the programmed feedrate.

Example

absolute G90

N.. G95

.....

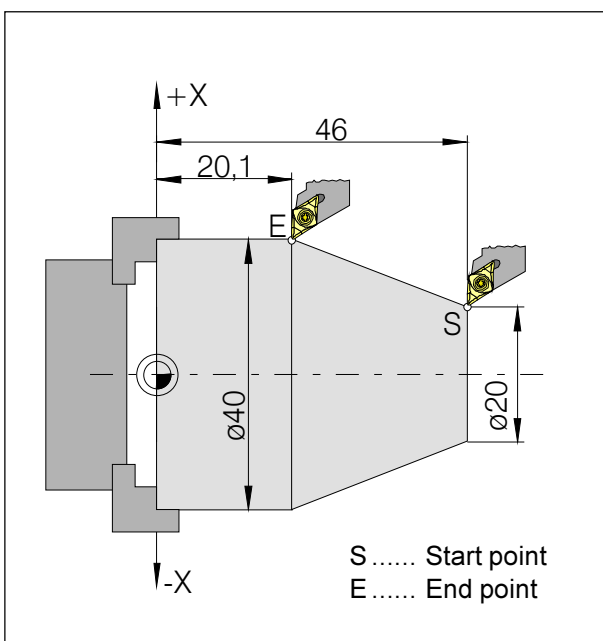
N20 G01 X40 Z20.1 F0.1

incremental G91

N.. G95 F0.1

.....

N20 G01 X20 W-25.9



Absolute and incremental measures for G00

Insertion of Chamfers and Radii

Example

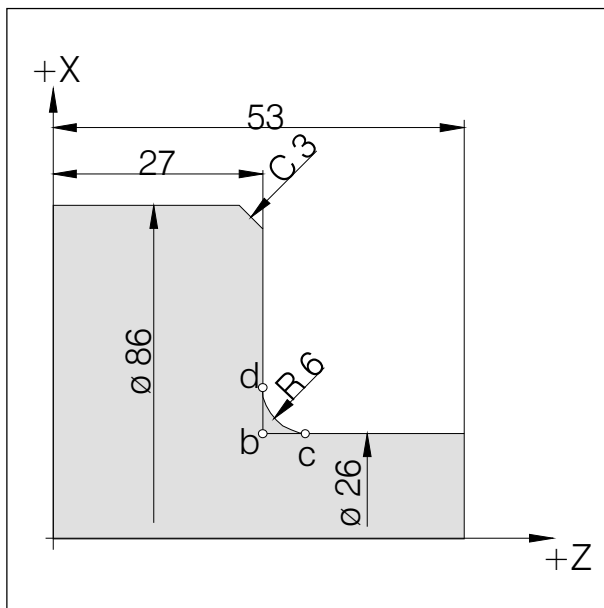
```

....
N 95   G 01   X 26   Z 53
N 100  G 01   X 26   Z 27   R 6
N 105  G 01   X 86   Z 27   C 3
N 110  G 01   X 86   Z 0
....

```

Notes

- Chamfers and radii can be inserted between two G00/G01 movements only.
- The movement, which is programmed in the second block, has to start at point b (drawing). With incremental programming the distance from point b has to be programmed.
- With single block mode the tool stops first at point c and then at point d.
- If the movements in one of the blocks are so short, that there is with inserting a chamfer or radius no intersection point, alarm no. 055 occurs.



Insertion of chamfers and radii

Direct draw input

	Commands	Tool movements
1	$X_2... (Z_2...) A...$	
2	$A_1... X_3... Z_3... A_2...$	
3	$X_2... Z_2... R... X_3... Z_3... A_2...$ oder $A_1... R... X_3... Z_3... A_2...$	
4	$X_2... Z_2... C... X_3... Z_3... A_2...$ oder $A_1... C... X_3... Z_3... A_2...$	
5	$X_2... Z_2... R_1... X_3... Z_3... R_2... X_4... Z_4...$ oder $A_1... R_1... X_3... Z_3... A... R_2... X_4... Z_4...$	

	Commands	Tool movements
6	$X_1... Z_1... C_1...$ $X_3... Z_3... C_2...$ $X_4... Z_4...$ oder $A_1... C_1...$ $X_3... Z_3... A_2... C_2...$ $X_4... Z_4...$	
7	$X_2... Z_2... R_1...$ $X_3... Z_3... C_2...$ $X_4... Z_4...$ oder $A_1... R_1...$ $X_3... Z_3... A_2... C_2...$ $X_4... Z_4...$	
8	$X_2... Z_2... C_1...$ $X_3... Z_3... R_2...$ $X_4... Z_4...$ oder $A_1... C_1...$ $X_3... Z_3... A_2... R_2...$ $X_4... Z_4...$	

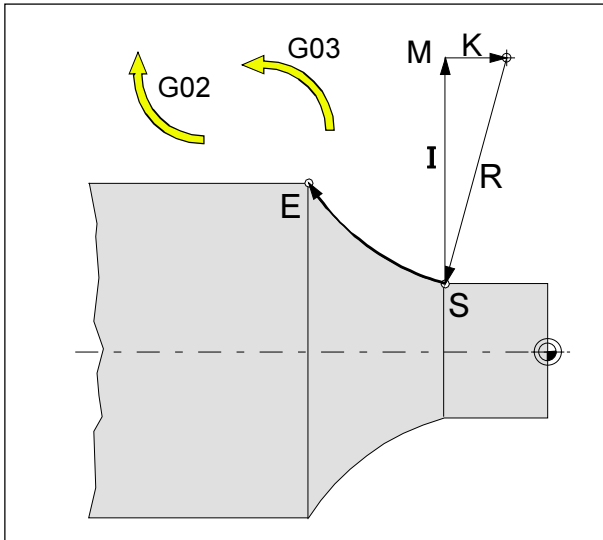
Missing intersection point coordinates need not to be calculated.

In programs angles (A), chamfers (C) and radii (R) can be programmed directly.

Note

The following G commands must not be used for the blocks with chamfer or radius. They must not be used between the blocks with chamfer or radius, which define the succession numbers.

- G-Codes (except G04) in group 00
- G02, G03, G20, G21 and G24 in group 01



Rotational direction and parameter of an arc

G02 Circular Interpolation Clockwise

G03 Circular Interpolation Counterclockwise

Format

N... G02 X(U)... Z(W)... I... K... F...

or

N... G02 X(U)... Z(W)... R... F...

X,Z U, W ... End point of the arc

I,K..... Incremental circle parameters

(Distance from start point to centre of arc, I is related to X, K to Z)

R..... Radius of arc

The tool will be traversed to the target point along the defined arc with the programmed feed F.

Note

- Programming the value 0 for I and K can be omitted.
- Input of R with a positive sign effects an arc $<180^\circ$, a negative sign effects an arc $>180^\circ$.
- A full circle can not be programmed with R.
- The observation of G02, G03 occurs always behind the turning axis (according to DIN 66 025), wherever machining occurs in front of or behind the turning axis.

G04 Dwell

Format

N... G04 X(U)... [sec]

or

N... G04 P... [msec]

The tool movement will be stopped at the last reached position for a dwell defined by X, U or P - sharp edges - transitions, cleaning cut-in ground, precise stop

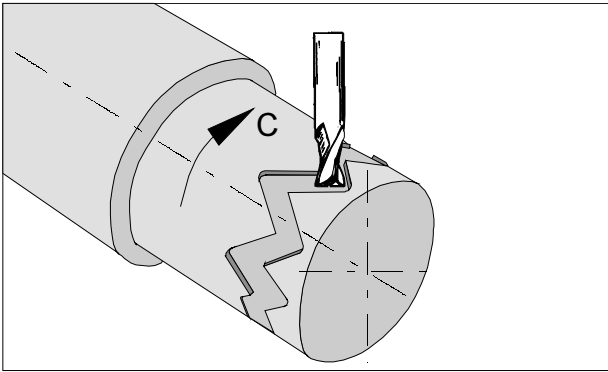
Note

- With address P no decimal point is allowed
- The dwell time starts at the moment when the tool movement speed is zero.
- t max. = 2000 sec, t min. = 0,1 sec
- input resolution 100 msec (0,1 sec)

Examples

N75 G04 X2.5 (dwell time= 2.5sec)

N95 G04 P1000 (dwell time = 1 sec = 1000 msec)



G7.1 Cylindrical Interpolation

Format:

N... G7.1 C...

N... G7.1 C0

G7.1 C... Starts the cylinder interpolation.
The C- value describes the radius of the the blank part.

G7.1 C0 End of cylinder interpolation

The tool tip position 0 must be programmed for all tools that will be used for the cylindrical interpolation.

This function enables the development of a cylinder surface in programming.

In this way e.g. programs for cylindrical cam machining on lathes can be created.

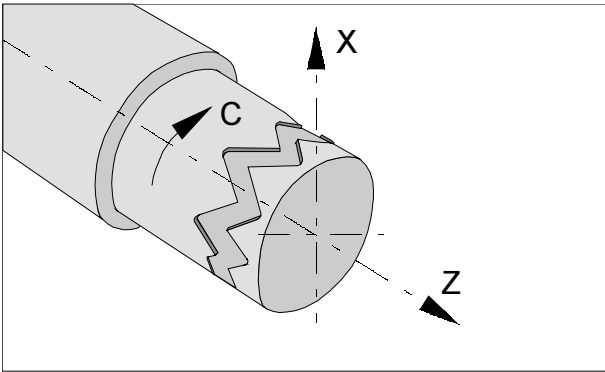
The traverse amount of the rotary axis C programmed by indication of the angle is converted in the control into the distance of a fictitious linear axis along the external surface of the cylinder.

Thus, it is possible that linear and circular interpolations on this area can be carried out with another axis.

With G19 the level is determined in which the rotary axis C is preset in parallel to the Y-axis.

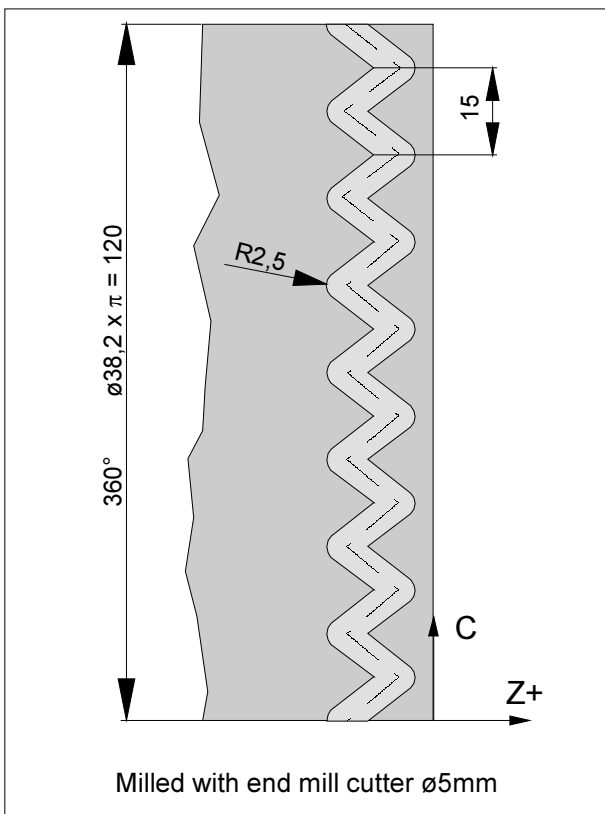
Notes:

- The reference point of the cylinder must be entered incrementally, since otherwise it would be approached by the tool!
- In the offset data cutter position 0 must be allocated to the tool. However, the miller radius must be entered.
- In mode G7.1 the coordinate system must not be changed.
- G7.1 C.. and/or G13.1 C0 must be programmed in the mode "cutter radius compensation off" (G40) and cannot be started or terminated within "cutter radius compensation on" (G41 or G42).
- G7.1 C.. and G7.1 C0 must be programmed in separate blocks.
- In a block between G7.1 C.. and G7.1 C0 an interrupted program cannot be restarted.
- The arc radius with circular interpolation (G2 or G3) must be programmed via an R-command and must not be programmed in degree and/or via K and J-coordinates.
- In the geometry program between G7.1 C.. and G7.1 C0 no rapid motion (G0) and/or positioning procedures causing rapid motion movements (G28) or drilling cycles (G83 to G89) must be programmed.
- The feed entered in the mode cylindric interpolation is to be considered as traverse speed on the unrolled cylinder area.



Example - Cylindrical Interpolation

X axis with diametrical programming and C axis with angular programming.



O0002 (Cylindrical Interpol.)

N15 T0505

N25 M13 Sense of rotation for driven tools
(be equivalent to M3)

N30 G97 S2000

N32 M52 Positioning of the spindle

N35 G7.1 C19.1 Start of the interpolation /
blank part radius

N37 G94 F200

N40 G0 X45 Z-5

N45 G1 X35 C0 Z-5

N50 G1 Z-15 C22.5

N55 Z-5 C45

N60 Z-15 C67.5

N65 Z-5 C90

N70 Z-15 C112.5

N75 Z-5 C135

N80 Z-15 C157.5

N85 Z-5 C180

N90 Z-15 C202.5

N95 Z-5 C225

N100 Z-15 C247.5

N105 Z-5 C270

N110 Z-15 C292.5

N115 Z-5 C315

N120 Z-15 C337.5

N125 Z-5 C360

N130 X45

N135 G7.1 C0 End of interpolation
N140 M53 Ende des roundaxis
operation

N145 G0 X80 Z100 M15

N150 M30

G10 Data Setting

The command G10 allows to overwrite control data, programming parameters, writing tool data etc...
G10 is frequently used to program the workpiece zero point.

Example: Zero point offset

Format

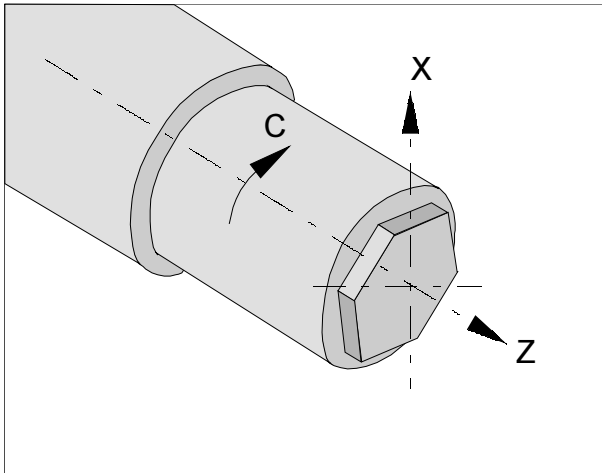
N... G10 P...X...Z...R...Q...;
oder
N... G10 P...U...W...C...Q...;

P: wear offset number

0	Traverse value for the workpiece-coordinate system
1-64	Tool tear correction value The Comand value is the offset number
10000+(1-64)	tool geometry offset number (1-64)

X...	Offset number in the X- axis (asolute)
Z...	Offset number in the Z- axis (absolute)
U...	Offset number in the X- axis (inkremental)
W...	Offset number in the Z- axis (inkremental)
R...	tool nose radius offset value (absolute)
R...	tool nose radius offset value (inkrementel)
Q...	imaginary tool nose number

With G10 P0 the workpiece zero point become overwrite.
For this reason the work piece lenght etc can be taken into consideration.



The tool tip position 0 must be programmed for all tools that will be used for the **polar coordinate interpolation**.

G12.1/G13.1 Polar Coordinate Interpolation

Format:

N... G12.1

N... G13.1

G12.1 Starts the polar coordinate interpolation

G13.1 Terminates the polar coordinate interpolation

The polar coordinate interpolation is adequate for machining the end face of a turned part.

It converts a command programmed in the Cartesian coordinate system into the movement of a linear axis X (tool movement) and a rotating axis C (workpiece rotation) for the path control.

Notes:

- Also with diameter programming for the linear axis (X-axis) radius programming is used for the rotary axis (C-axis).
- Cutter position 0 must be assigned to the tool in the offset data.
However, the miller radius must be entered.
- In G12.1- mode the coordinate system must not be altered.
- G12.1 and/or G13.1 must be programmed in the mode "cutter radius compensation off" (G40) and cannot be started or terminated within "cutter radius compensation on" (G41 or G42).
- G12.1 and G13.1 are to be programmed in separate blocks.
In a block between G12.1 and G13.1 an interrupted program cannot be brought to a new start.
- The arc radius with circular interpolation (G2 oder G3) can be programmed by means of an R-command and/or via I- and J-coordinates.
- In the geometry program between G12.1 and G13.1 no rapid motion (G0) must be programmed (see adjoining scheme).

G12.1 chooses a level (G17) in which the polar coordinate interpolation is carried out.

The level G18 used by G12.1 before programming is deleted.

It is reestablished by means of the command G13.1 (polar coordinate interpolation end).

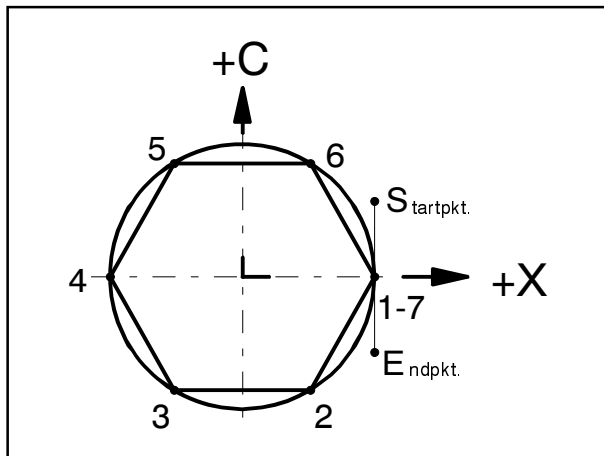
After switching on the machine or with a system RESET the condition "polar coordinate interpolation" is also cancelled, (G13.1) and the level defined by means of G18 is used.

G-codes which may be programmed in the mode "polar coordinate interpolation:

G-Code	Use
G01	Linear interpolation
G02, G03	Circular interpolation
G04	Intermission
G40, G41, G42	Cutter radius compensation (polar coordinate interpolation is applied on the tool path after the tool compensation)
G65, G66, G67	User macro command
G98, G99	Feed per minute, feed per rotation

Example - Polar Coordinate Interpolation

X axis with diametrical programming and C axis with angular programming.



Point	X	C
S	34,64	10
1	34,64	0
2	17,32	-15
3	-17,32	-15
4	-34,64	0
5	-17,32	15
6	17,32	15
7	34,64	0
E	34,64	-10

O0003 (Polarc. Interpol.)

N5 T0303

N10 M13 Sense of rotation for
 driven tools
 (be equivalent to M3)

N15 G97 S2000

N20 M52

Positioning of the spindle

N25 G12.1

Start of the interpolation

N30 G0 X60 Z-6 C10

N35 X34.64 C10 G41

N45 G1 C0 F0.2

N50 X17.32 C-15

N55 X-17.32 C-15

N60 X-34.64 C0

N65 X-17.32 C15

N70 X17.32 C15

N75 X34.64 C0

N80 C-10

N85 G1 X45 C-10 G40

N90 G13.1

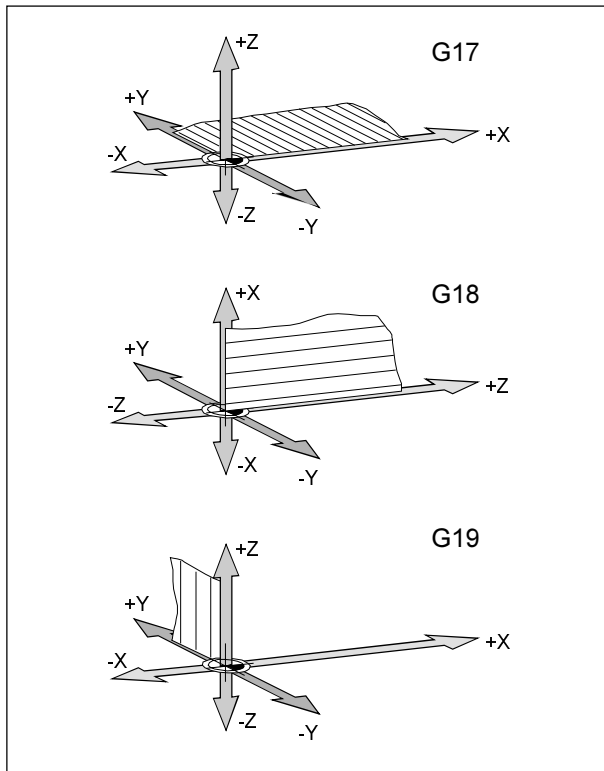
End of interpolation

N95 M53

Ende des roundaxisoperation

N100 G0 X80 Z100 M15

N105 M30



Definition of the main planes

G17-G19 Plane Selection

Format

N... G17/G18/G19

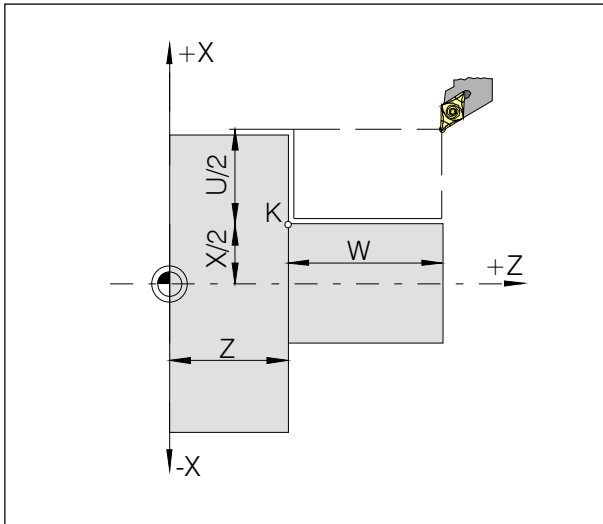
With G17 to G19 the plane will be defined, in which circular interpolation and polar coordinate interpolation can be proceeded and in which the cutter radius compensation will be calculated.

In the vertical axis to the active plane the tool length compensation will be proceeded.

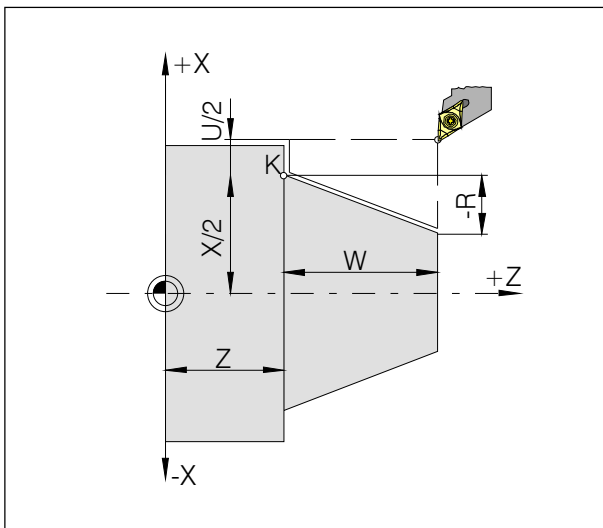
G17 XY-Plane

G18 ZX-Plane

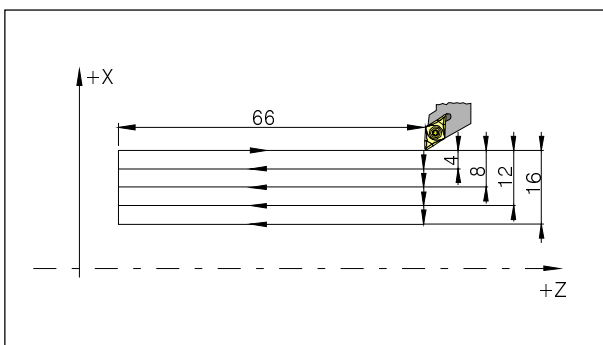
G19 YZ-Plane



Longitudinal turning cycle without taper



Longitudinal turning cycle with negative taper R



Example: G20 Longitudinal turning cycle

G20 Longitudinal Turning Cycle

Format

N... G20 X(U)... Z(W)... F... (straight)

or

N... G20 X(U)... Z(W)... R... F... (taper)

X(U), Z(W) Absolute (incremental) coordinates of the contour point K

R [mm] Incremental taper dimension in X with direction (+/-)

Notes

- This cycle is modal and will be deselected by a G command of the same group.
- For following blocks only the altered coordinates have to be programmed (see example).
- A negative taper parameter (-R) defines the taper as shown in the drawing.

N100 G91

.....

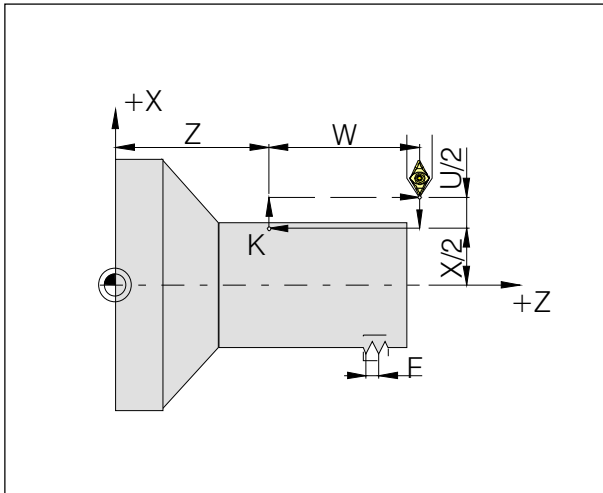
N110 G20 U-4 W-66 F0.18

N115 U-8

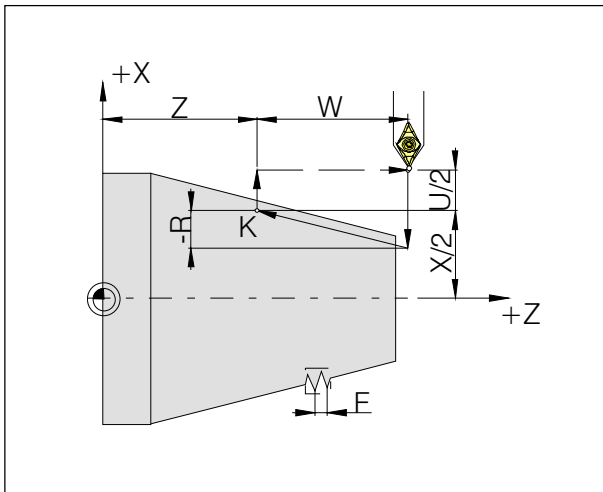
N120 U-12

N125 U-16

N130 G00



Straight thread cutting cycle



Tapered thread cutting cycle

G21 Thread Cutting Cycle

Format

N... G21 X(U)... Z(W)... F... (straight)

or

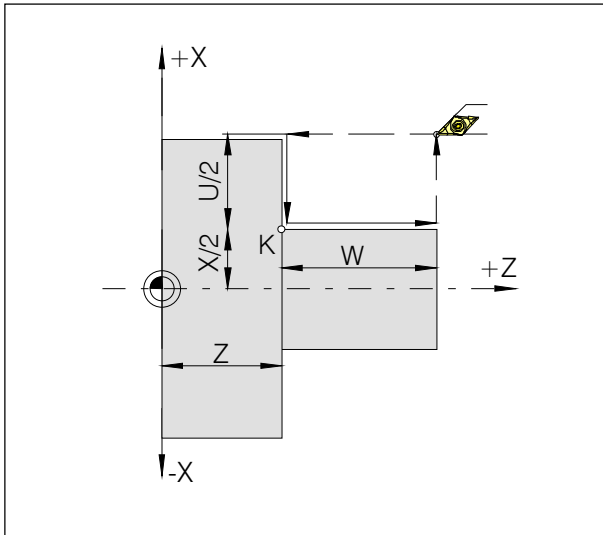
N... G21 X(U)... Z(W)... R... F... (taper)

F Thread pitch [mm]

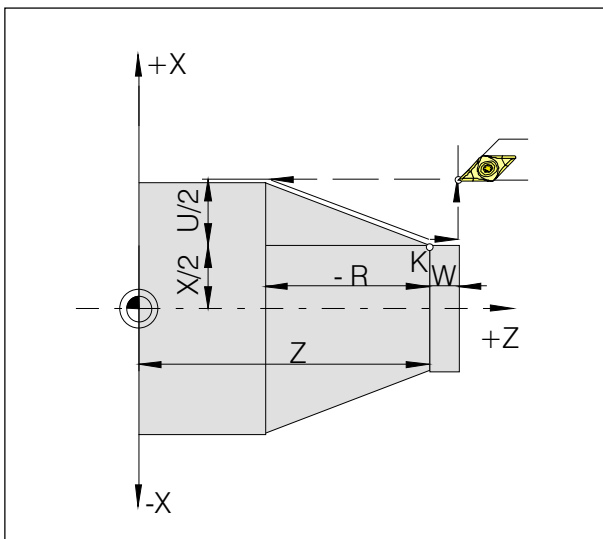
R [mm] Incremental taper dimension in X
with direction (+/-)

Notes

- This cycle is modal and will be deselected by a G command of the same group.
- For following blocks only the altered coordinates have to be programmed (see example).
- A negative taper parameter (-R) defines the taper as shown in the drawing.



Face turning cycle without taper



Face turning cycle with negative taper R

G24 Face Turning Cycle

Format

N... G24 X(U)... Z(W)... F... (straight)

or

N... G24 X(U)... Z(W)... R... F... (taper)

R..... Incremental value of the taper in Z axis

Notes

- This cycle is modal and will be deselected by a G command of the same group.
- For following blocks only the altered coordinates have to be programmed (see example).
- A negative taper parameter (-R) defines the taper as shown in the drawing.

G28 Return to Reference Point

Format

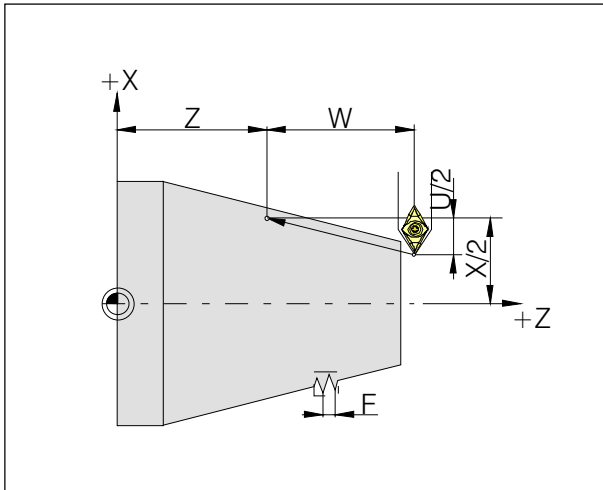
N... G28 X(U)... Z(W)...

X,Z..... Absolute intermediate coordinates

U,W ... Incremental intermediate coordinates

The G28 command is used to approach the reference point via an intermediate position (X(U), Z(W)). First the movement to X(U) and Z(W), then movement to the reference point. Both movements with G00!

Zero point shift G92 will be deselected.



Measures for thread cutting

G33 Thread Cutting

Format

N... G33 X(U)... Z(W)... F...

F Thread pitch [mm]

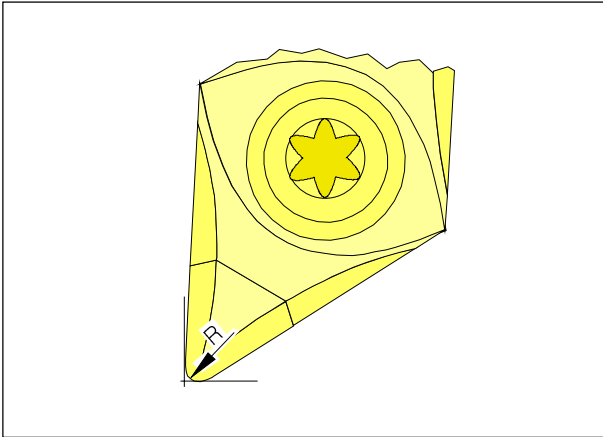
Straight, tapered and scroll threads can be cut.

Because of no automatic return to the start point, the multiple threading cycle G78 will be preferred.

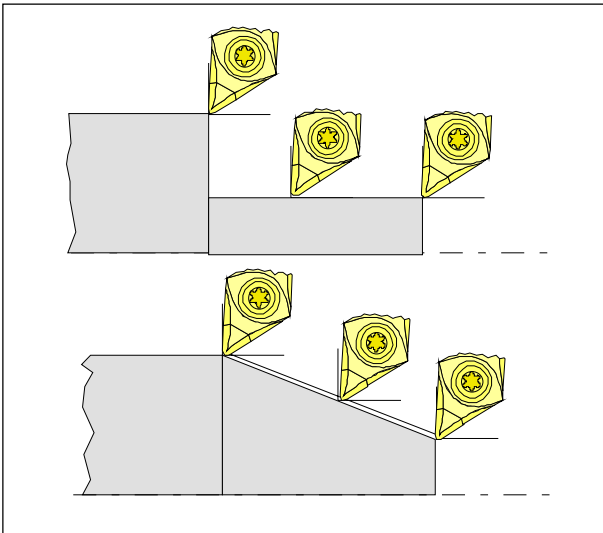
Machining routines like knurling are also possible.

Note

- With taper threading the thread pitch has to be defined with the higher value in X or Z axis.
- Continuous thread cutting is possible (multiple threads)



Tool tip radius and theoretical cutter tip



Movements parallel to the axes and oblique

Cutter Radius Compensation

During tool measurement the tool tip is measured only at two points (touching the X and Z axes).

The tool offset therefore only describes a theoretical cutter tip.

This point is traversed on the workpiece in the programmed paths.

With movements in the axis directions (longitudinal and face turning) the points on the tool tip touching the axes are used.

No dimensional errors are therefore produced on the workpiece.

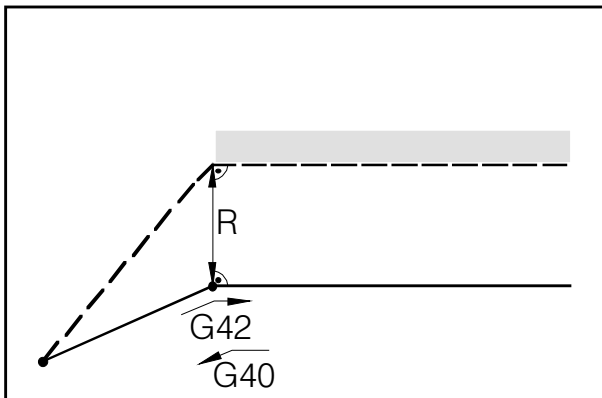
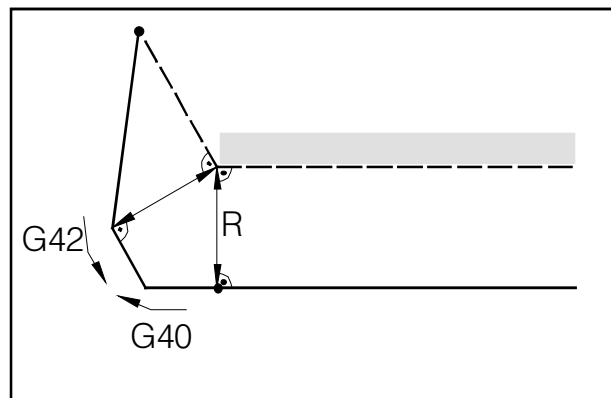
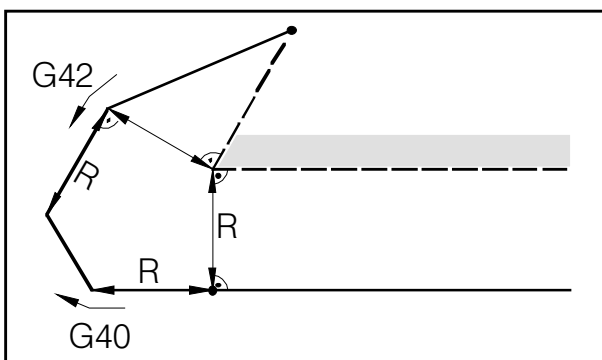
With simultaneous movements in both axis directions (tapers, radii) the position of the theoretical cutter point no longer coincides with the point on the tool tip actually cutting.

Dimensional errors occur on the workpiece.

Maximum dimensional error without cutter radius compensation with 45° movements:

Tool tip radius $0,4 \text{ mm} \pm 0,16 \text{ mm}$ path distance $\pm 0,24 \text{ mm}$ distance in X and Z.

If cutter radius compensation is used, these dimensional errors are automatically calculated and compensated by the control.

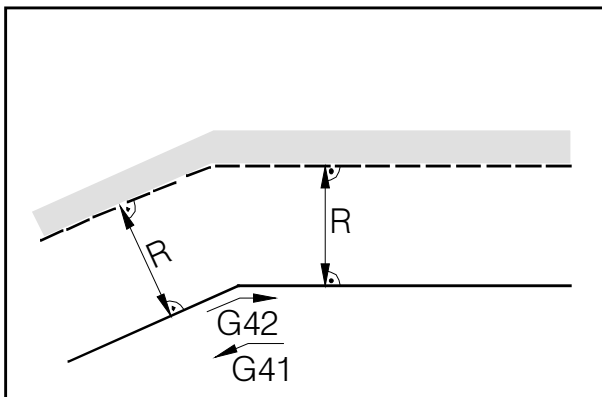
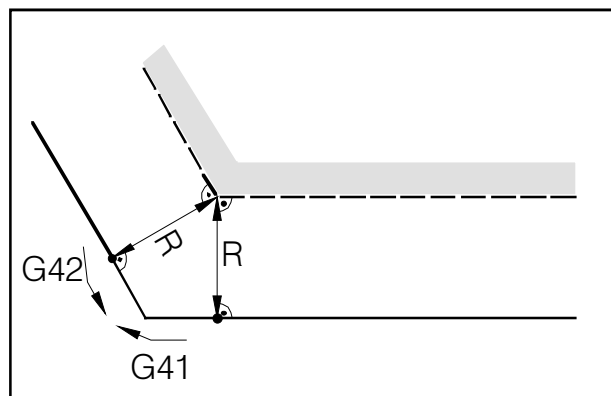
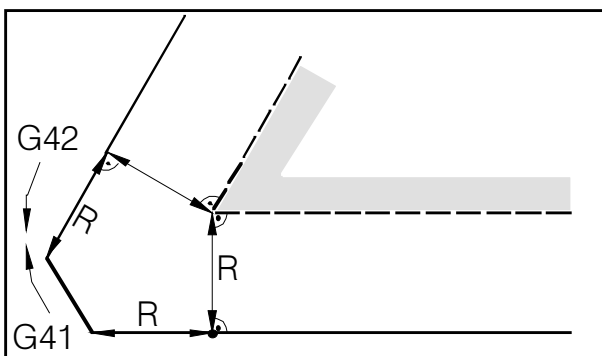
Tool pathes with selection / cancellation of the cutter radius compensation*Frontal approach or leaving of an edge point**Approach or leaving an edge point at side behind**Approach or leaving an edge point behind*

— — — programmed tool path
 ————— real traversed tool path

With arcs always the tangent of the end or start point of the arc will be approached.

The approaching path to the contour and the leaving path from the contour must be larger than the tool tip radius R , otherwise program interruption with alarm.

If contour elements are smaller than the tool tip radius R , contour violations could happen. The software computes three blocks forward to recognize this contour violations and interrupt the program with an alarm.

Tool pathes with program run with active cutter radius compensation*Tool path at an inner edge**Tool path at an outer edge > 90°**Tool path at an outer edge < 90°*

— — — programmed tool path
 ————— real traversed tool path

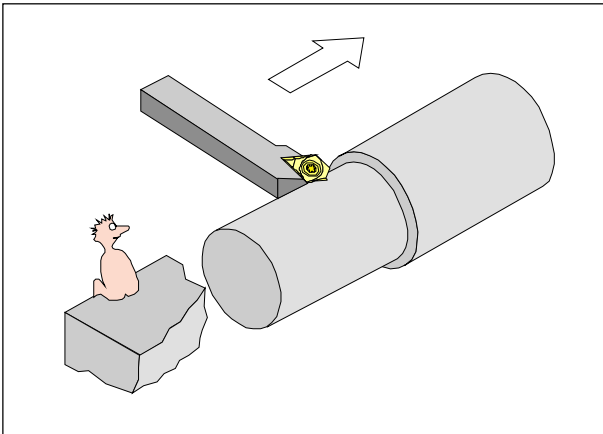
With arcs always the tangent of the end or start point of the arc will be approached.

If contour elements are smaller than the tool tip radius R , contour violations could happen. The software computes three blocks forward to recognize this contour violations and interrupt the program with an alarm.

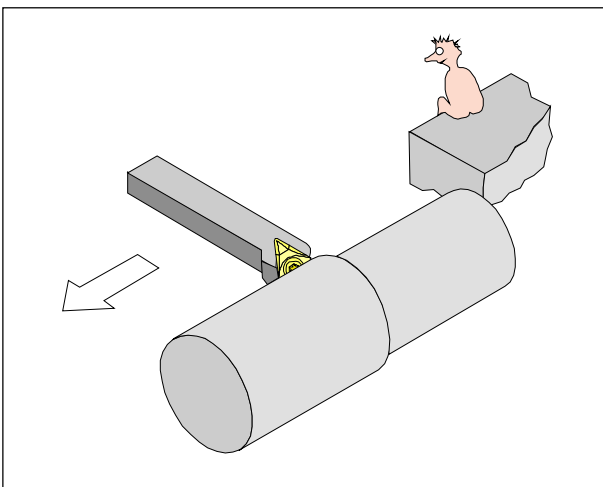
G40 Cancel Cutter Radius Compensation

The radius compensation will be cancelled with G40. Cancellation is only permitted in combination with a linear traversing command (G00, G01)

G00 or G01 can be programmed in the same block or as the first traversing movement after cancellation.



Definition G41 cutter radius compensation left



Definition G42 cutter radius compensation right

G41 Cutter Radius Compensation Left

If the programmed tool path (viewed in the direction of machining) is on the left of the material to be machined, the radius compensation is to be selected with G41.

Notes

- No direct change between G41 and G42 - cancel with G40 previously.
- Cutter radius R and cutter position T must be defined.
- Selection is only permitted in conjunction with G00 or G01.
- Change of tool correction is not possible with active cutter radius compensation.

G42 Cutter Radius Compensation Right

If the programmed tool path (viewed in the direction of machining) is on the right of the material to be machined, the radius compensation is to be selected with G42.

Notes see G41!

G70 Measuring in Inches

Format

N5 G70

By programming G70 the following values will be converted to inches:

- Feedrate F [mm/min, inch/min, mm/rev, inch/rev]
- Offset values (zero point, geom., wear, ...) [mm, inch]
- Movement pathes [mm, inch]
- Display of actual position [mm, inch]
- Speed [m/min, feet/min]

Notes

- For clearness G70 should be defined in the first block of the program
- The measuring system which was programmed last will be active - also after main power off/on.
- To get back to the origin measuring system it is best to use the MDI mode (e.g. MDI G70 Cycle Start)

G71 Metrical Measuring

Format

N5 G71

See G70!

G72 Finishing Cycle

Format

N... G72 P... Q...

P..... Block number of the first block for the program of finishing shape.

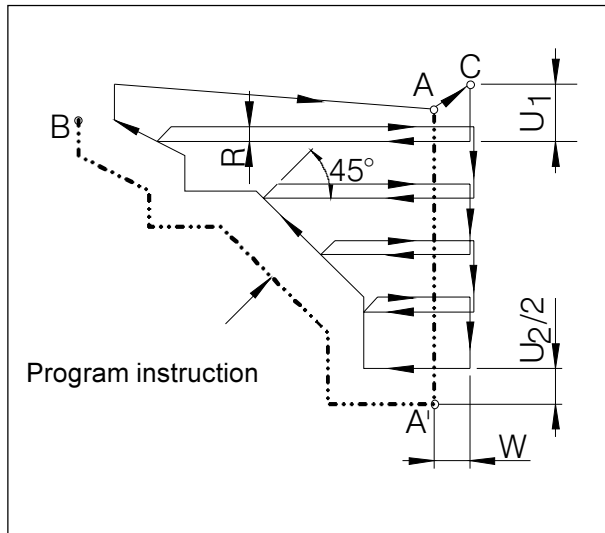
Q Block number of the last block for the program of finishing shape.

After rough cutting by G73, G74, G75 the G72 command permits finishing.

The programmed shape between P and Q which was also used for rough cutting, will be repeated without cutting depth division and without finishing offset.

Notes

- F, S and T functions and G41 or G42 specified between P and Q are only effective for G72. They are not effective for G73, G74 and G75!
- The finishing cycle G72 must only be programmed after the cycles G73, G74 and G75.



Turning cycle contour

Example:

Machining of the contour illustrated above. U_1 , R , U_2 and W (in N100 and N110) are very enlarged because of better showing.

A: $X=102, Z=0$

A': $X=20, Z=0$

B: $X=100, Z=-50$

Program:

```

...
N70 G00 X102 Z0                (point A)
...
...
N100 G73 U8 R2
N110 G73 P120 Q190 U10 W5
N120 G0 X20                    (point A')
N130 G1 Z-15
N140 X40
N150 X70 Z-30
N160 Z-40
N170 X80
N180 X90 Z-50
N190 X100                      (point B)
N200 S... F... T...            (select finishing tool)
N210 G72 P120 Q190            (finishing cycle)
....

```

G73 Turning Cycle Contour**Format**

N... G73 U_1 ... R ...

N... G73 P ... Q ... U_2 +/-... W +/-... F ... S ... T ...

first block U_1 [mm] depth of cut, incremental, without sign, in the drawing shown as U_1

R [mm] retract height

second block P block number of the first block for the programmed shape

Q block number of the last block for the programmed shape

U_2 [mm] distance and direction of finishing offset in X direction (diameter or radius designation), in the drawing shown as $U_2/2$

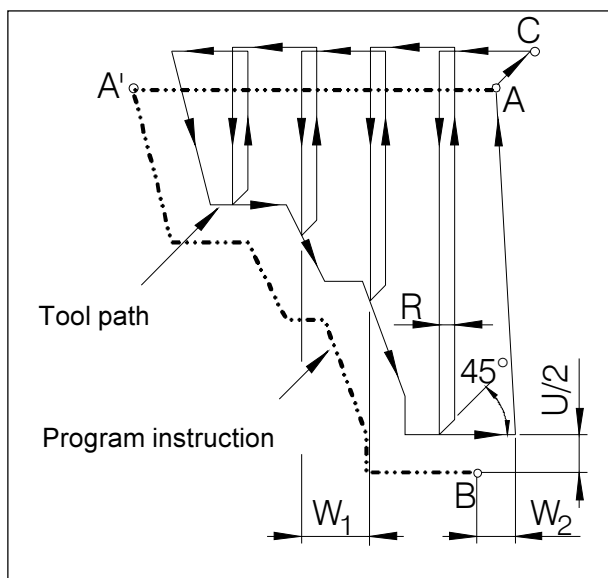
W [mm] Distance and direction of finishing offset in Z direction, incremental, without sign

F , S , T Feed, speed, tool

Before machining the tool is at point C. Between the block numbers P and Q a contour (A to A' to B) will be programmed, it will be machined with the corresponding cutting depth division onto the defined finishing offset (2. block, in the drawing $U_2/2$).

Notes

- F , S and T functions between P and Q are ignored.
- The contour between A' and B has to be programmed increasing, that means the diameter has to increase.
- The first movement from A to A' must be $G00$ or $G01$, is permitted in X only ($G00 X...$) and must be programmed in absolute coordinates.
- Between P and Q no subroutine call is permitted.



Facing cycle contour

Example:

Machining of the contour illustrated above. W_1 , R , W_2 and U (in N100 and N110) are very enlarged because of better showing.

A: X=120, Z=2

A': X=120, Z=-45

B: X=20, Z=0

Program:

```

...
N70 G00 X120 Z2                                (point A)
...
...
N100 G74 W9 R2
N110 G74 P120 Q190 U10 W5
N120 G0 Z-45                                    (point A')
N130 G1 X80 Z-40
N140 Z-30
N150 X60 Z-25
N160 Z-20
N170 X30 Z-15
N180 X20
N190 Z0                                          (point B)
N200 S... F... T...                            (select finishing tool)
N210 G72 P120 Q190                            (finishing cycle)
....

```

G74 Facing Cycle Contour

Format

N... G74 W_1 ... R ...

N... G74 P... Q... U+/-... W_2 +/-... F... S... T...

first block W_1 [mm] depth of cut in Z, incremental, without sign, in the drawing shown as W_1

R [mm] retract height

second block P..... block number of the first block for the programmed shape

Q block number of the last block for the programmed shape

U [mm] distance and direction of finishing offset in X direction (diameter or radius designation), in the drawing shown as $U/2$

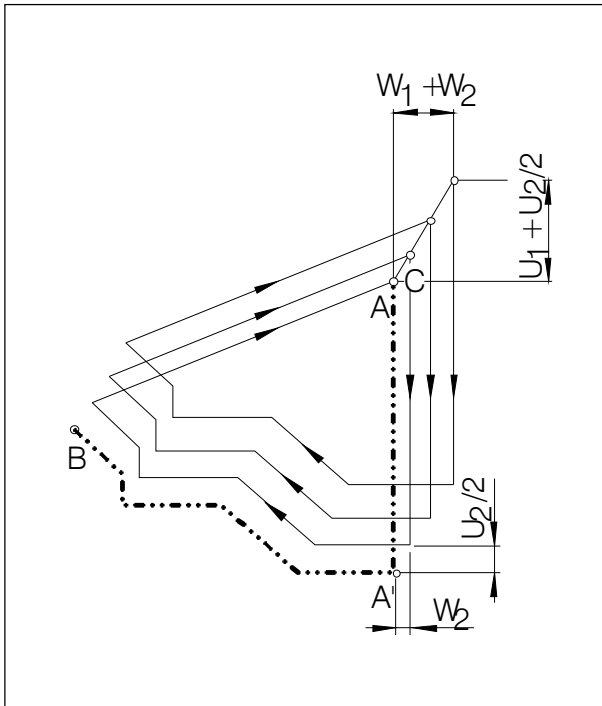
W_2 [mm] Distance and direction of finishing offset in Z direction, incremental, without sign, in the drawing shown as W_2

F, S, T Feed, speed, tool

Before machining the tool is at point C. Between the block numbers P and Q a contour (A to A' to B) will be programmed, it will be machined with the corresponding cutting depth division onto the defined finishing offset (2. block, in the drawing W_2).

Notes

- F, S and T functions between P and Q are ignored.
- The contour between A' and B has to be programmed decreasing, that means the diameter has to decrease.
- The first movement from A to A' must be G00 or G01, is permitted in Z only (G00 X...) and must be programmed in absolute coordinates.
- Between P and Q no subroutine call is permitted.



Pattern repeating

G75 Pattern Repeating

Format

N... G75 U₁+/-... W₁+/-... R...

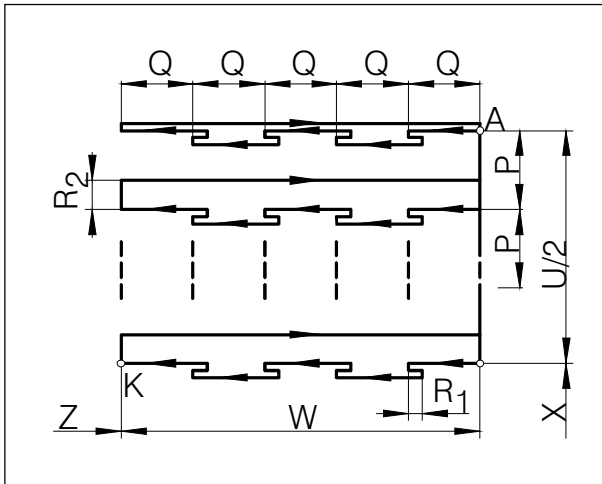
N... G75 P... Q... U₂... W₂... F... S... T...

- first block U₁ Start point for the cycle in the X axis (radius designation), in the drawing shown as U₁
- W₁ Start point of the cycle in the Z axis, in the drawing shown as W₂
- R number of repetitions (equal to cut division)
- second block P block number of the first block for the programmed shape
- Q block number of the last block for the programmed shape
- U₂ [mm] distance and direction of finishing offset in X direction (diameter or radius designation), in the drawing shown as U/2
- W₂ [mm] Distance and direction of finishing offset in Z direction, incremental, without sign, in the drawing shown as W₂
- F, S, T Feed, speed, tool

The G75 cycle allows machining parallel to the shape of the workpiece, the pattern will be shifted to the finished shape step by step.

Application for semifinished products (forged, cast parts)

G76 Deep Hole Drilling, Cut-In Cycle in Z



Deep hole drilling, cut-in cycle in Z

Format

N... G76 R...

N... G76 X(U)... Z(W)... P... Q... R... F...

first block R [mm] retraction height for chip breaking (incremental without sign), drawing: R₁

sec. block X(U), Z(W) Absolute (incremental) coordinates of the contour edge point K

or

Z(W) Absolute (incremental) drilling depth

P [μm] Incremental feed in X direction (no sign); P < tool width!

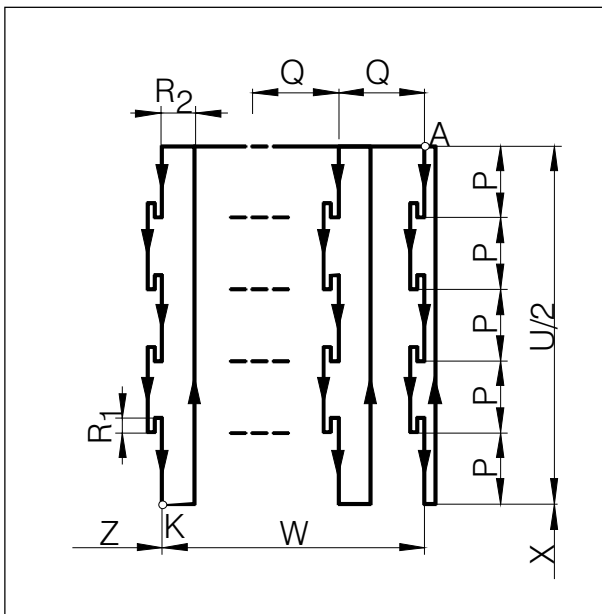
Q [μm] Cutting depth in Z direction (no sign)

R Undercut at Z end point, drawing: R₂

F Feed rate

Notes

- Without addresses X(U) and P G76 can be used as drilling cycle (Move tool to X=0 before!)
- With cut-in cycle the infeed P has to be smaller than tool width B.
- With the first cut no undercut will be executed at the Z end point.
- The undercut must be a positive value.



Cut-in cycle in X

G77 Cut-In Cycle in X

Format

N... G77 R...

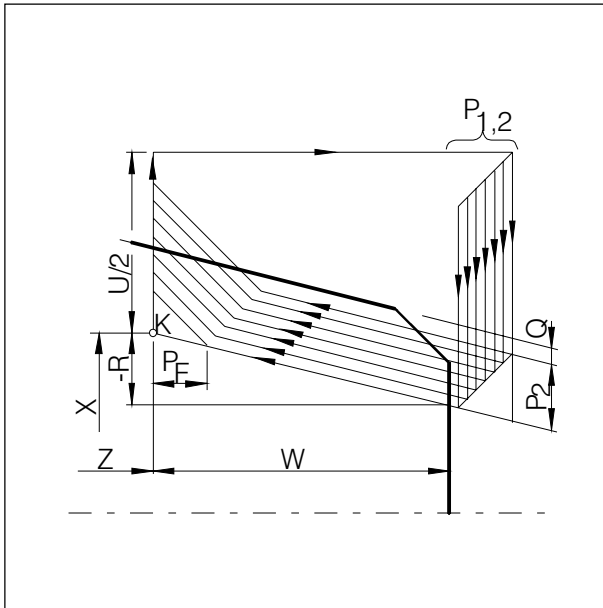
N... G77 X(U)... Z(W)... P... Q... R... F...

first block	R [mm]	Retraction height for chip breaking, in the drawing shown as R_1
sec. block	X(U), Z(W)	Absolute (incremental) coordinates of K
	P [μ m]	Cutting depth in X direction (no sign)
	Q [μ m]	Incremental infeed in Z direction (no sign)
	R	Undercut at X end point, in the drawing shown as R_2
	F	Feed

Note

- The infeed Q must be smaller than tool width B.
- Tool width will not be taken into consideration with this cycle.
- With the first cut no undercut will be executed.
- The undercut must be a positive value.

G78 Multiple Threading Cycle



Multiple threading cycle

Format

N... G78 P... Q... R...

N... G78 X(U)... Z(W)... R... P... Q... F...

first block P..... is a 6 digit parameter divided in 2 digit groups

- PXXxxxx
Defines the number of finishing cuts

- PxxXXxx
Defines the chamfer value (see P_F in drawing) [mm]
 $P_F = F \times P_{xxXXxx} / 10$

- PxxxxXX
Defines the flank angle [°]
allowed: 80, 60, 55, 30, 29, 0

Q Minimum cutting depth [μ m]
incremental

R Finishing offset [mm]
incremental

second block X(U), Z(W) Absolute (incremental) coordinates of the point K

R [mm] Incremental taper value with sign (R=0 straight thread)

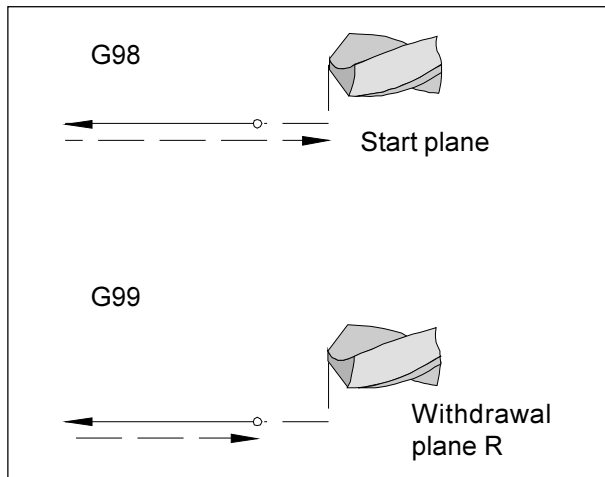
P [μ m] Thread depth (always positive), in the drawing shown as P_2

Q [μ m] Cutting depth of the first cut (radius value) without sign

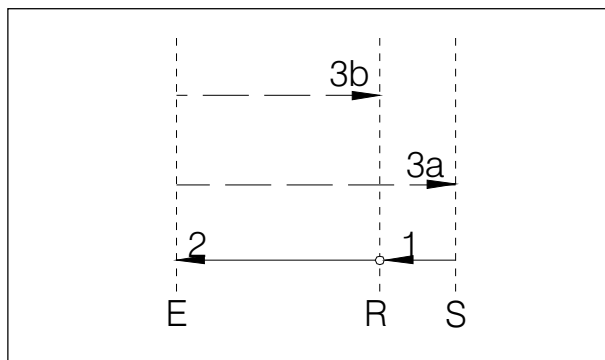
F [mm] Thread pitch

Note

Negative taper parameter R defines the taper as shown in the drawing.



Retraction characteristics G98, G99



Sequence of movements G98, G99

Systematic G98/G99

G98 After reaching the drilling depth the tool retracts to the starting plane

G99 After reaching the drilling depth the tool retracts to the withdrawal plane - defined by the R parameter

If no G98 or G99 is active, the tool retracts to the starting plane. If G99 (retraction to withdrawal plane) is programmed, The address R must be defined. With G98 R need not to be programmed.

R defines the height of the withdrawal plane related to the last Z position (start position of the drilling cycle). With a negative value for R the withdrawal plane will be below the start position, with a positive value the withdrawal plane will be over the start position

Sequence of movements

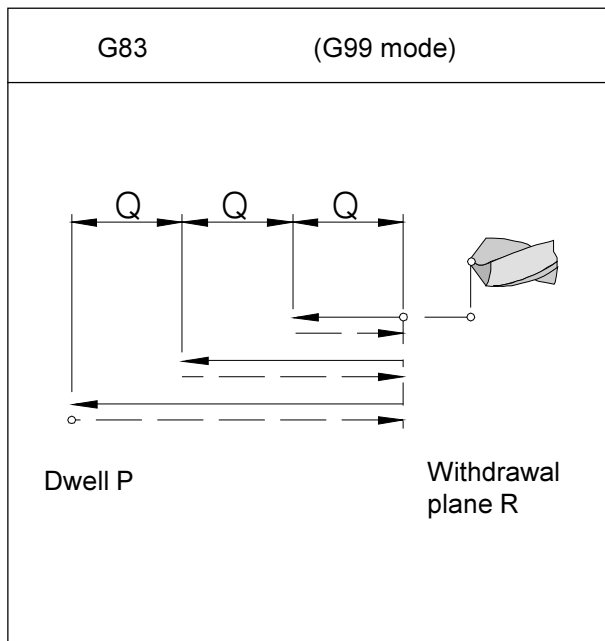
- 1: The tool traverses with rapid speed from the start position (S) to the plane defined by R (R).
- 2: Cycle-specific drill machining down to end depth (E).
- 3: The withdrawal occurs a: with G98 to the start plane (S) and b: with G99 to the withdrawal plane.

G80 Cancel Cycles G83 - G85

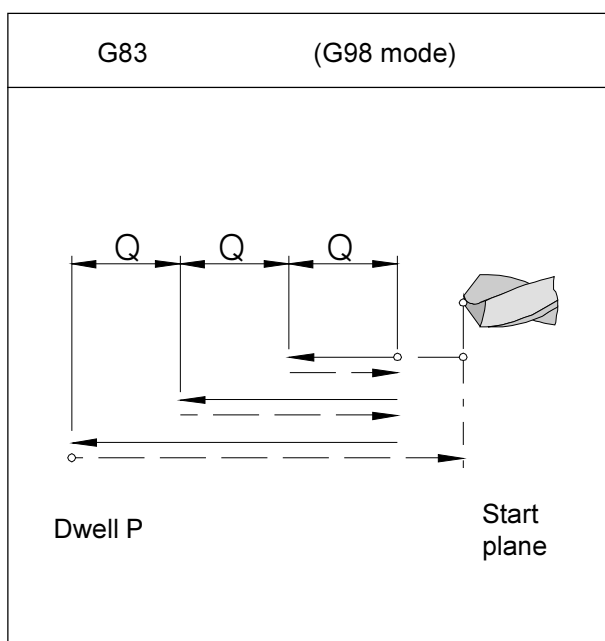
Format

N... G80

The drilling cycles are modal. They have to be cancelled by G80 or another group 1 command (G00, G01, ...).



Drilling cycle with retraction to withdrawal plane



Drilling cycle with retraction to start plane

G83 Drilling Cycle

Format

N... G98(G99) G83 X0 Z(W)... (R...) Q... P... F... M...

G98(G99) .. Return to starting plane (withdrawal plane)

X0 Hole position in X (always zero)

Z(W) Absolute (incremental) drilling depth

R [mm] Incremental value for the withdrawal plane related to the start point in Z (with sign)

Q [μm] Drilling depth per cut

P [msec] Dwell at the hole bottom

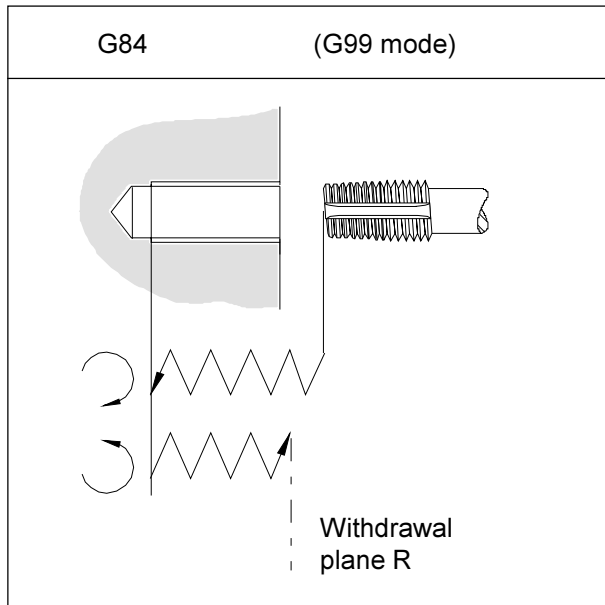
P1000 = 1 sec

F Feed rate

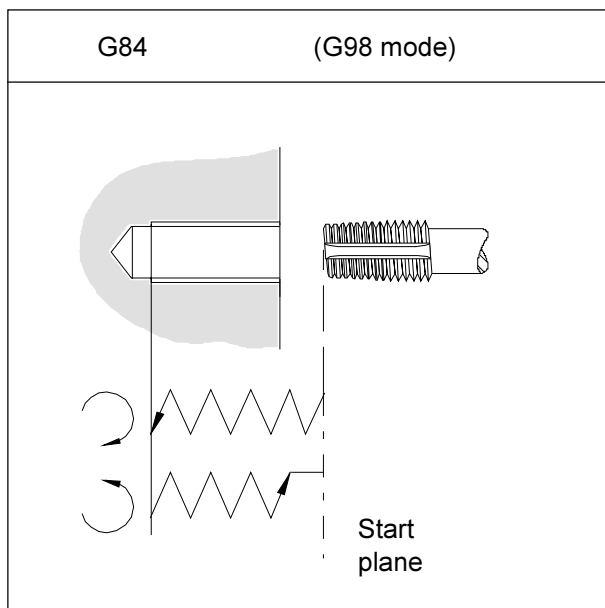
M Spindle direction (M03 or M04)

Notes

- Parameter R must be specified when G99 is programmed. You can leave R when G98 is active.
- It is possible to leave address X0 at G83 if you have programmed the starting position for execution of the cycle in both axis (N... G00 X0 Z...) at the block before.
Otherwise just the starting position in Z axis has to be approached (N.. G00 Z3) at the block before.
- When Q is not specified no cut division will be done, that means drilling until Z end point in one movement.



Tapping cycle with retraction to the withdrawal plane



Tapping cycle with retraction to the start plane

G84 Tapping Cycle

Format

N... G98(G99) G84 X0 Z(W)... (R...) F... M...

F Thread pitch

X0 Hole position in X (always zero)

Z(W) Absolute (incremental) drilling depth

R [mm] Incremental value for the withdrawal plane related to the start point in Z (with sign)

P [msec] Dwell at the hole bottom

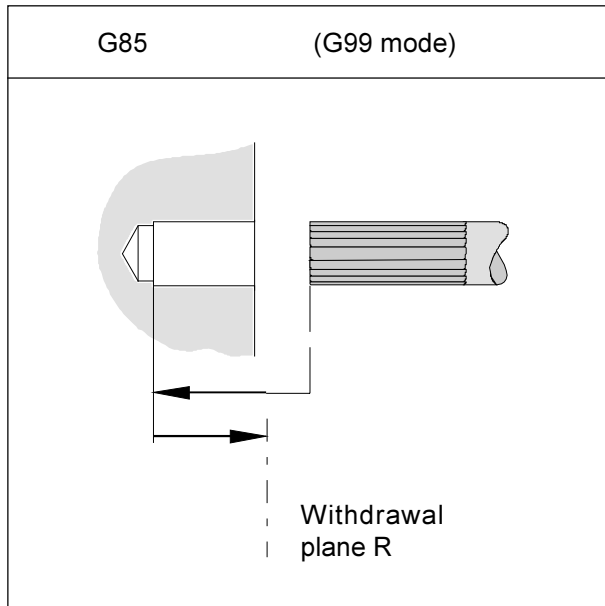
P1000 = 1 sec

F Feed rate

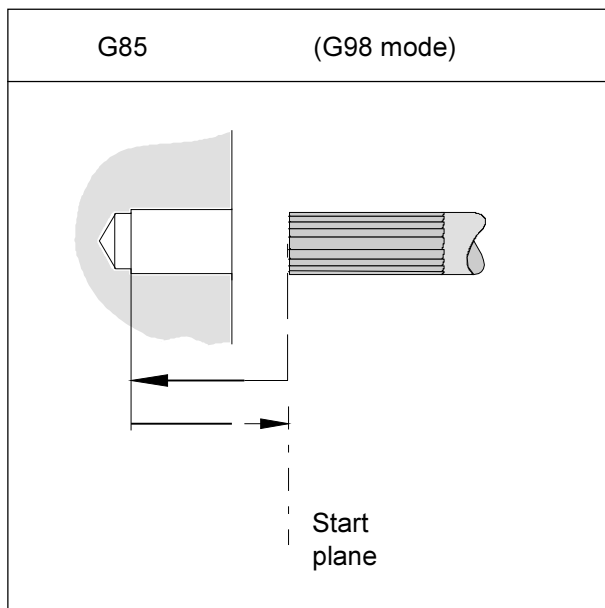
M Spindle direction (M03 or M04)

Notes

- Parameter R must be specified when G99 is programmed. You can leave R when G98 is active.
- It is possible to leave address X0 at G83 if you have programmed the starting position for execution of the cycle in both axis (N... G00 X0 Z...) at the block before. Otherwise just the starting position in Z axis has to be approached (N.. G00 Z3) at the block before.
- Start the tapping cycle with the adequate spindle direction (M03 or M04). At the tapping ground the spindle direction changes automatically for retraction. After the tapping cycle the spindle direction changes to the original direction.



Reaming cycle with retraction to the withdrawal plane



Reaming cycle with retraction to the start plane

G85 Reaming Cycle

Format

N... G98(G99) G85 X0 Z(W)... (R...) P... F... M...

X0 Hole position in X (always zero)

Z(W) Absolute (incremental) drilling depth

R [mm] Incremental value for the withdrawal plane related to the start point in Z (with sign)

P [msec] Dwell at the hole bottom

P1000 = 1 sec

F Feed rate

M Spindle direction (M03 or M04)

Notes

- Parameter R must be specified when G99 is programmed. You can leave R when G98 is active.
- It is possible to leave address X0 at G83 if you have programmed the starting position for execution of the cycle in both axis (N... G00 X0 Z...) at the block before.
Otherwise just the starting position in Z axis has to be approached (N.. G00 Z3) at the block before.
- Start the tapping cycle with the adequate spindle direction (M03 or M04). At the tapping ground the spindle direction changes automatically for retraction. After the tapping cycle the spindle direction changes to the original direction.

Retraction to the start point is executed with double the programmed feed.

A cut division by defining Q is not possible.

G90 Absolute Programming

Format

N... G90

The addresses have to be programmed as following:

X Diameter

U+/- Incremental in diameter (expect some cycles)

Z+/- Absolute (referred to the workpiece zero point)

W+/- ... Incremental (actual) traverse distance

Notes

- Direct switchover from block to block between G90 and G91 is permitted.
- G90 and G91 may also be programmed with some other G functions (N... G90 G00 X... Z...).

G91 Incremental Programming

Format

N... G91

The addresses have to be programmed as following:

X,U Diameter

Z,W Incremental (actual) traverse distance with sign

Notes see G90.

G92 Spindle Speed Limit

The max. spindle speed for a part program can be established with the command G92. The desired value is entered under the address S (spindle).

This command is used in conjunction with programmed cutting speed.

No further commands are allowed in the same block.

G92 Coordinate System Setting

Format

N... G92 X... Z... (Set coordinate system)

or

N... G92 U... W... (Shift coordinate system)

or

N... G92 S... (Spindle speed limitation)

Sometimes it is necessary to shift the workpiece zero point in a program. This is not possible for the adjustable zero offset. Therefore G92 is a possibility to handle this problem.

Example

You want to shift your workpiece zero from the right to the left side of the workpiece

Diameter of workpiece = 30 mm

Length of workpiece = 100 mm

Program

```
N... G90 ..... Programming absolute
... ..... Workpiece zero point right
... ..... Right side of contour is
           finished
N180 G00 X35 ..... Retract
N185 Z-100 ..... Movement distance = work-
           piece length
N190 G92 X35 Z0 .. New zero point on the left side
... ..... Workpiece zero point left
... ..... Machine left side
N305 G00 X35 ..... Retract
N310 Z100 ..... Movement distance = work-
           piece length
N315 G92 X35 Z0 .. Workpiece zero point left
           again
... ..... etc.
```

Zero offset with G92 is modal and it is not cancelled through M30 or RESET!

In this way, do not forget to reset the zero offset G92 before program is finished.

When zero offset will be inserted incrementally, the values U and W will be added to the last valid zero offset.

G94 Feed Rate in Minutes

The entry of the command G94 means that all values programmed under "F" (feed) are in mm/min.

G95 Feed Rate in Revolutions

The entry of the command G95 means that all commands programmed under "F" are in mm/revolution.

G96 Constant Cutting Speed

Unit: m/min

The control continually computes the spindle speed corresponding to the respective diameter.

G97 Constant Rotational Speed

Unit: rev/min

Description of M Commands

M Commands are switching or additional functions (miscellaneous). The M commands can stand alone in a program block or together with other commands. Commands of the same group cancel each other, that means the M command programmed last cancels the previously programmed M command of the same group.

Remark

The following pages describe the standard M commands. Whether these M commands are executable depends on the type of the machine and the used accessories.

M00 Programmed Stop Unconditional

These command effects a stop in the execution of the part program.

Main spindle, feed and coolant will be switched off. The chip protection door can be opened without triggering an alarm.

With "NC START"  the program run can be continued. After that the main drive will be switched on with all values which were valid before.

M01 Programmed Stop Conditional

M01 works like M00, but only if the function PROGRAMMED STOP YES was switched on by softkey in the menu PROGRAM CONTROL.

With "NC START"  the program run can be continued. After that the main drive will be switched on with all values which were valid before.

M02 Main Program End

M02 works like M30.

M03 Main Spindle ON Clockwise

The spindle is switched on provided that a spindle speed or cutting speed has been programmed, the chip protection door is closed and a workpiece correctly clamped.

M03 must be used for all right-hand cutting or overhead clamped tools, if machining occurs behind the turning centre.

M04 Main Spindle ON Counterclockwise

The same conditions as described under M03 apply here.

M03 must be used for all left-hand cutting or normal clamped tools, if machining occurs behind the turning centre.

M05 Main Spindle Off

The main drive is braked electrically.

At the program end the main spindle is automatically switched off.

M08 Coolant ON

only for EMCO PC Turn 120/125/155.
The coolant will be switched on.

M09 Coolant OFF

only for EMCO PC Turn 120/125/155.
The coolant will be switched off.

M17 Subroutine End

M17 will be written in the last block of a subroutine. It can stand alone in this block or with other functions. The call-up of a subroutine and M17 must not stand in the same block (nesting).

M20 Tailstock BACK

only for accessory automatical tailstock.
The tailstock moves back.
See "Accessory Functions - Automatic Tailstock"

M21 Tailstock FORWARD

only for accessory automatical tailstock only on the PC TURN 120/125/155.
The tailstock moves forward.
See "Accessory Functions - Automatic Tailstock"

M25 Open Clamping Device

only for accessory automatical clamping device only on the PC TURN 120/125/155.
The clamping device opens.
See "Accessory Functions - Automatic Clamping Device"

M26 Close Clamping Device

only for accessory automatical clamping device only on the PC TURN 120/125/155.
The clamping device closes.
See "Accessory Functions - Automatic Clamping Device"

M30 Program End

With M30 all drives are switched off and the control is returned to the start of the program.
Moreover, the counter level is increased by 1.

M71 Puff Blowing ON

only for accessory puff blowing device
The puff blowing device will be switched on.

M72 Puff Blowing OFF

only for accessory puff blowing device
The puff blowing device will be switched off.

M98 Subprogram Call

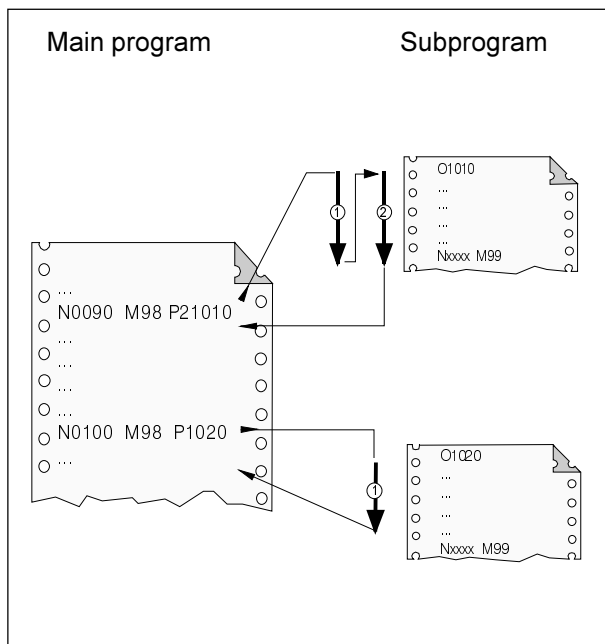
Format

N... M98 P...

P The first four digits from the right determine the subprogram number, the other digits the number of repetitions.

Notes

- M98 can be designated in the same block as the movement command (e.g. G01 X25 M98 P25001).
- When the count of repetitions is not specified, the subprogram is called once (M98 P5001).
- When the programmed subroutine does not exist an alarm occurs.
- A two loop subprogram call can be executed.



Subroutine call from a main program

M99 Subprogram End, Jump Instruction

Format

N... M99 P...

M99 in the main program

Without jumping address:

Jump to the program start.

With jumping address Pxxxx:

Jump on block no. xxxx

M99 in the subprogram

Without jumping address:

Jump to the calling up program, on the next block after the calling up block (see drawing).

With jumping address Pxxxx:

Jump to the calling up program on block no. xxxx

Note

- M99 must be the last command in the subprogram.
- The jump-back occurs automatically into the next following block in the main program.

G: Flexible NC programming

Variable number	Variable type	Function
#0	Always zero system variable	This variable has always the value zero. Not changeable
#1-33	Local variable	At disposal for calculations in the program
#100-149	Global variables	At disposal for calculations in the program
#500-531	System variable	At disposal for calculations in the program
#1000	System variable	Loading magazine: bar end reached
#1001	System variable	Loading magazine: loader has advanced
#1002	System variable	Loading magazine: first part after bar change
#3901	System variable	Nominal piece number
#3901	System variable	Actual piece number

Variables and arithmetic parameters

By using variables instead of fixed values, a program can be configured more flexibly. Thus, you can react to signals, such as e.g. measuring values, or the same program can be used for different geometries by using variables as nominal value.

Together with variable calculation and program jumps you get the possibility to create a highly-flexible program archive and thus save programming time.

Local and global variables can be read and written. All other variables can only be read.

Local variables can only be used in that macro in which they are defined.

Global variables can be used in every macro irrespective of the macro in which they were defined.

Calculating with variables

With the four basic arithmetic operations the usual mathematic notation is valid.

The term at the operator's right can contain constants and/or variables combined by functions.

Each variable can be replaced again by an arithmetic term in square brackets or by a constant.

Example

```
#1=#[#2]
```

During the calculation the limitation is valid that the execution of the calculation is carried out from left to right without observance of the calculation rule point before line.

Example

```
#1=#2*3+#5/2
```

Function	Example
=	#1=2
+	#1=#2+#3
-	#1=#2-#3
*	#1=#2*#3
/	#1=#2/#3

Control structures

In programs the control sequence can be changed by IF and GOTO instructions. Three types of branchings are possible:

- IF[<condition>] THEN
- IF[<condition>] GOTO <n>
- GOTO <destination>

IF[<Condition>] THEN

After IF a provisory term must be indicated. If the provisory term applies, a determined macro instruction is carried out. Only one macro instruction can be carried out.

Example

With equal values of #1 and #2 the value 5 is allocated to #3.

```
IF [#1 EQ #2] THEN#3=5
```

IF[<Condition>] GOTO <n>

After IF a provisory term must be indicated. If the provisory term applies, the branching is carried out to block number n. Otherwise the subsequent block is carried out.

Example

If the value of the variable #1 is greater than 10, the branching is carried out to block number N4. Otherwise the subsequent block is carried out.

```
IF [#1 GT 10] GOTO 4
```

GOTO <n>

The jump command GOTO can also be programmed without condition. A variable or constant can be used as a branch destination. With a variable the number can be replaced again by a calculation term in square brackets.

Example

Jump to block number 3

```
GOTO 3
```

Example

Jump to variable #6

```
GOTO#6
```

Relational operators

Relational operators consist of two letters and are used to determine, in comparison with two values, if these are equal or if one value is greater and/or less than the other.

Operator	Meaning
EQ	Equal (=)
NE	Unequal (≠)
GT	Greater than (>)
GE	Greater than or equal (≥)
LT	Less than (<)
LE	Less than or equal (≤)

The expressions to be compared can be variable n or constants. A variable can be replaced again by a calculation term in square brackets.

Example

```
IF[#12 EQ 1] GOTO10
```

Comprising macro programming examples:

```
IF[#1000 EQ 1] GOTO10
```

```
IF[#10] NE #0] GOTO#1
```

```
IF[1 EQ 1] THEN#2 =5
```

```
IF[#4+#2/2] GT #20] THEN#10] =#1*5+#7
```

H: Alarms and Messages

Missing digitizer calibration

Cause: A digitizer tablet has been installed but not calibrated

Remedy: Calibrate digitizer tablet (set corner points), see External Input Devices

6: CONVERTER ALREADY INITIALIZED

System error. Re-install the software.

7: MISSING SETUP CALL

System error. Re-install the software.

8: SETUP OF PROGRAM CONTROLLING FAILS

System error. Re-install the software.

9: PARSER SETUP FAILS

System error. Re-install the software.

10: REGISTRY SETUP FAILS

System error. Re-install the software.

11: SETUP OF WORKING POINTS FAILS

System error. Re-install the software.

12: SETUP OF WORKING OBJECTS FAILS

System error. Re-install the software.

13: SETUP OF COMMAND LIST FAILS

System error. Re-install the software.

14: SETUP OF START CONDITION FAILS

System error. Re-install the software.

15: SETUP OF EXPORT VARIABLE FAILS

System error. Re-install the software.

16: SETUP OF MAIN VARIABLE FAILS

System error. Re-install the software.

17: WAITING FOR AC INITIALIZATION

System error. Re-install the software.

18: SETUP OF AC FAILS

System error. Re-install the software.

19: INVALID SWITCHTONEXTBLOCK ID

System error. Re-install the software.

20: NO PROGRAM

System error. Re-install the software.

21: PROGRAM NOT FOUND

System error. Re-install the software.

1000: PARSER - ERROR OT_FIRST

System error. Re-install the software.

1001: MARKING OF HEADER LINE NOT FOUND

System error. Re-install the software.

1002: MARKING OF MAIN PROGRAM NOT FOUND

System error. Re-install the software.

1005: INVALID LINE NUMBER

System error. Re-install the software.

1006: NO END OF FUNCTION FOUND

System error. Re-install the software.

1007: MODUL NAME ALREADY EXISTS

System error. Re-install the software.

1009: INVALID MODUL NAME

System error. Re-install the software.

1010: NO LINE NUMBER

System error. Re-install the software.

1018: WAIT FOR F OR S COMMAND

NC programming error. G4 was programmed without S or F address.

1020: NO MORE COMMANDS ALLOWED

NC programming error. G96 must be the only G command in a block.

1035: NO OR INVALID PARAMETER

System error. Re-install the software.

2016: INVALID S VALUE

NC programming error. Spindle index invalid. Allowed indexes are S[0] and S[1].

Input Device Alarms 3000 - 3999

These Alarms will be triggered by the control keyboard or digitizer.

Missing digitizer calibration

Cause: A digitizer tablet has been installed but not calibrated

Remedy: Calibrate digitizer tablet (set corner points), see External Input Devices

3001 General RS232 communication error

Remedy: Correct settings of serial interface.

3002 Control keyboard missing

Remedy: Connect control keyboard, switch on, ...

3003 Digitizer missing

Remedy: Connect digitizer, switch on, ...

3004 Check sum error in control keyboard

The keyboard tries an automatic re-initializing - when failed switch off / on keyboard.

3005 Error in control keyboard

The keyboard tries an automatic re-initializing - when failed switch off / on keyboard.

3006 Error with initializing control keyboard

The keyboard tries an automatic re-initializing - when failed switch off / on keyboard.

Machine Alarms 6000 - 7999

These alarms will be triggered by the machines. There are different alarms for the different machines. The alarms 6000 - 6999 normally must be confirmed with RESET. The alarms 7000 - 7999 are messages which normally will disappear when the releasing situation is finished.

PC MILL 50 / 55, PC TURN 50 / 55

The following alarms are valid for the turning and milling machines of the series 50 / 55.

6000: EMERGENCY OFF

The EMERGENCY OFF key was pressed. Remove the endangering situation and restart machine and software.

6001: CYCLE TIME EXCEEDS LIMIT

Contact EMCO Service.

6002: NO PLC PROGRAM LOADED

Contact EMCO Service.

6003: DB NOT EXISTENT

Contact EMCO Service.

6004: RAM ERROR ON PLC BOARD

Contact EMCO Service.

6009: FAILURE SAFETY CIRCUIT

Defective door limit switch or main contactor. Operating the machine is not possible.
Contact EMCO Service.

6010: X-AXIS NOT READY

Step motor board defective, 24 V or 30 V fuse defective. Check fuses and switch box fan filter.
Contact EMCO Service.

6011: Y-AXIS NOT READY

see alarm 6010.

6012: Z-AXIS NOT READY

see alarm 6010.

6013: MAIN DRIVE NOT READY

Main drive power supply defective, cable defective, main drive overload.
Check fuse, reduce load.
Contact EMCO service.

6014: NO SPEED FOR MAIN SPINDLE

This will be released, when the spindle speed is lower than 20 rpm because of overload.
Alter cutting data (feed, infeed, spindle speed).

6019: VICE TIMEOUT

24 V fuse defective, hardware defective.
Contact EMCO service.

6020: VICE FAILURE

24 V fuse defective, hardware defective.
Contact EMCO service.

6024: DOOR NOT CLOSED

The door was opened while a machine movement.
The program will be aborted.

6025: GEARBOX COVER NOT CLOSED

The gearbox cover was opened while a machine movement. A running CNC program will be aborted.
Close the cover to continue.

6027: DOOR LIMIT SWITCH DEFECTIVE

The limit switch of the automatic door is displaced, defective, wrong cabled.
Contact EMCO service.

6028: DOOR TIMEOUT

The automatic door sticks, the pressured air supply is insufficient, the limit switch is displaced.
Check door, pressured air supply, limit switch or contact EMCO service.

6030: NO PART CLAMPED

No workpiece inserted, vice cheek displaced, control cam displaced, hardware defective.
Adjust or contact EMCO service.

6031: QUILL FAILURE**6037: CHUCK TIMEOUT****6039: CHUCK PRESSURE FAILURE****6041: TOOL CHANGE TIMEOUT**

Tool turret sticks (collision?), 24 V fuse defective, hardware defective.

A running CNC program will be stopped.

Check for a collision or contact EMCO service.

6042: TOOL CHANGE TIMEOUT

see alarm 6041.

6043: TOOL CHANGE TIMEOUT

see alarm 6041.

6044: TOOL TURRET SYNC ERROR

Hardware defective.

Contact EMCO service.

6046: TOOL TURRET SYNC MISSING

Hardware defective.

Contact EMCO service.

6048: DIVIDING TIME EXCEEDED

Dividing head sticks, insufficient pressured air supply, hardware defective.

Check for collision, check pressured air supply or contact EMCO service.

6049: INTERLOCKING TIME EXCEEDED

see alarm 6048

6050: FAILURE DIVIDING DEVICE

Hardware defective.

Contact EMCO service.

7000: INVALID TOOL NUMBER

The CNC program will be stopped.

Interrupt program with RESET and correct the program.

7007: FEED HOLD

In the robotic mode a HIGH signal is at input E3.7. Feed Stop is active until a low signal is at E3.7.

7017: GO FOR REFERENCE POINT

Approach the reference point.

7040: DOOR OPEN

The main drive can not be switched on and NC-Start can not be activated.

Some accessories can be operated only with open machine door.

Close the machine to run a program.

7043: PIECE COUNT REACHED

A predetermined number of program runs was reached. NC-Start is locked. Reset the counter to continue.

7050: NO PART CLAMPED

After switching on or after an the vice is neither at the open position nor at the closed position.

NC-Start is locked.

Traverse the vice manually on a valid end position.

7051: DIVIDING DEVICE NOT INTERLOCKED

After switching on or after an the dividing head is not in a lock position. NC-Start is locked.

PC MILL 100/105/125/155

The following alarms are valid for the milling machines PC MILL 100/105/125/155.

6000: EMERGENCY OFF

The EMERGENCY OFF key was pressed. Remove the endangering situation and restart machine and software.

6001: PLC-CYCLE TIME EXCEEDING

Contact EMCO Service.

6002: PLC - NO PROGRAM CHARGED

Contact EMCO Service.

6003: PLC - NO DATA UNIT

Contact EMCO Service.

6004: PLC - RAM MEMORY FAILURE

Contact EMCO Service.

6005: OVERHEAT BRAKEMODUL

Main drive was braked too often, large changes of speed within a short time. E4.2 active

6006: OVERLOAD BRAKE RESISTOR

see 6005

6007: SAFETY CIRCUIT FAULT

Axis and main drive contactor with machine switched off not disabled. Contactor got stuck or contact error. E4.7 was not active during switch-on.

6009: SAFETY CIRCUIT FAULT

Defective step motor system.

A running CNC program will be interrupted, the auxiliary drives will be stopped, the reference position will be lost.

Contact EMCO Service.

6010: DRIVE X-AXIS NOT READY

The step motor board is defective or too hot, a fuse or cabling is defective.

A running program will be stopped, the auxiliary drives will be switched off, the reference position will be lost.

Check fuses or contact EMCO service.

6011: DRIVE Y-AXIS NOT READY

see alarm 6010.

6012: DRIVE Z-AXIS NOT READY

see alarm 6010.

6013: MAIN DRIVE NOT READY

Main drive power supply defective, main drive too hot, fuse defective.

A running program will be stopped, the auxiliary drives will be switched off.

Check fuses or contact EMCO Service.

6014: NO MAIN SPINDLE SPEED

This will be released, when the spindle speed is lower than 20 rpm because of overload.

Alter cutting data (feed, infeed, spindle speed).

The CNC program will be aborted, the auxiliary drives will be stopped.

6024: MACHINE DOOR OPEN

The door was opened while a machine movement. The program will be aborted.

6040: TOOL TURRET INDEX FAILURE

After WZW procedure drum pressed down by Z-axis. Spindle position wrong or mechanical defect. E4.3=0 in lower state

6041: TOOL CHANGE TIMEOUT

Tool drum stuck (collision?), main drive not ready, fuse defective, hardware defective.

A running CNC program will be stopped.

Check for collisions, check fuses or contact EMCO service.

6043-6046: TOOL DISK POSITION FAULT

Position error of main drive, error of position supervising (inductive proximity switch defective or disadjusted, drum allowance), fuse defective, hardware defective.

The Z axis could have been slipped out of the toothing while the machine was switched off.

A running CNC program will be stopped.

Contact EMCO service.

6047: TOOL DISK UNLOCKED

Tool drum turned out of locked position, inductive proximity switch defective or disadjusted, fuse defective, hardware defective.

A running CNC program will be interrupted.

Contact EMCO service.

When the tool drum is turned out of locked position (no defect), act as following:

Turn the drum into locking position manually

Change into MANUAL (JOG) mode.

Turn the key switch. Traverse the Z slide upwards, until the alarm disappears.

6050: M25 AT RUNNING MAIN SPINDLE

Cause: Programming mistake in NC program.

A running program will be aborted.

The auxiliary drives will be switched off.

Remedy: Correct NC program

6064: DOOR AUTOMATIC NOT READY

Cause: pressure failure automatic door
automatic door sticks mechanically
limit switch for open end position defective
security print circuits defect
cabling defective
fuses defective

A running program will be aborted.

The auxiliary drives will be switched off.

Remedy: service automatic door

6069: CLAMPING FOR TANI NOT OPEN

When opening the clamping pressure switch does not fall within 400ms. Pressure switch defective or mechanical problem. E22.3

6070: PRESSURE SWITCH FOR TANI MISSING

When closing the clamping pressure switch does not respond. No compressed air or mechanical problem. E22.3

6071: DIVIDING DEVICE NOT READY

Servo Ready Signal from frequency converter missing. Excess temperature drive TANI or frequency converter not ready for operation.

6072: VICE NOT READY

Attempt to start the spindle with an open vice or without clamped workpiece.

Vice sticks mechanically, insufficient compressed air supply, compressed air switch defective, fuse defective, hardware defective.

Check the fuses or contact EMCO service.

6073: DIVIDING DEVICE NOT READY

Cause: locking switch defective
cabling defective
fuses defective

A running program will be aborted.

The auxiliary drives will be switched off.

Remedy: service automatic dividing device
lock the dividing device

6074: DIVIDING TIME EXCEEDED

Cause: dividing device sticks mechanically
locking switch defective
cabling defective
fuses defective

A running program will be aborted.

The auxiliary drives will be switched off.

Remedy: service automatic dividing device

6075: M27 AT RUNNING MAIN SPINDLE

Cause: Programming mistake in NC program.

A running program will be aborted.

The auxiliary drives will be switched off.

Remedy: Correct NC program

7000: INVALID TOOL NUMBER PROGRAMMED

The tool position was programmed larger than 10.

The CNC program will be stopped.

Interrupt program with RESET and correct the program.

7016: SWITCH ON AUXILIARY DRIVES

The auxiliary drives are off. Press the AUX ON key for at least 0.5 sec. (to avoid accidentally switching on) to switch on the auxiliary drives.

7017: REFERENCE MACHINE

Approach the reference point.

When the reference point is not active, manual movements are possible only with key switch at position "setting operation".

7018: TURN KEY SWITCH

With NC-Start the key switch was in position "setting operation".

NC-Start is locked.

Turn the key switch in the position "automatic" to run a program.

7020: SPECIAL OPERATION MODE ACTIVE

Special operation mode: The machine door is opened, the auxiliary drives are switched on, the key switch is in position "setting operation" and the consent key is pressed.

Manual traversing the axes is possible with open door. Swivelling the tool turret is not possible with open door. Running a CNC program is possible only with standing spindle (DRYRUN) and SINGLE block operation.

For safety: If the consent key is pressed for more than 40 sec. the function of this key is interrupted, the consent key must be released and pressed again.

7021: INITIALIZE TOOL TURRET

The tool turret operating was interrupted.

No traversing operation is possible.

Press tool turret key in JOG operation. Message occurs after alarm 6040.

7022: INITIALIZE TOOL TURRET !

see 7021

7038: LUBRICATION SYSTEM FAULT

The pressure switch is defective or gagged.

NC-Start is locked. This can be reset only by switching off and on the machine.

Contact EMCO service.

7039: LUBRICATION SYSTEM FAULT

Not enough lubricant, the pressure switch is defective.

NC-Start is locked.

Check the lubricant and lubricate manually or contact EMCO service.

7040: MACHINE DOOR OPEN

The main drive can not be switched on and NC-Start can not be activated (except special operation mode)

Close the machine to run a program.

7042: INITIALIZE MACHINE DOOR

Every movement and NC-Start are locked.

Open and close the machine door to initialize the safety circuits.

7043: PIECE COUNT REACHED

A predetermined number of program runs was reached. NC-Start is locked. Reset the counter to continue.

7054: VICE OPEN

Cause: the workpiece is not clamped

When switching on the main spindle with M3/M4 alarm 6073 (vice not ready) will be released.

Remedy: Clamp

7055: DIVIDING DEVICE NOT LOCKED

Cause: the dividing device is not locked

When switching on the main spindle with M3/M4 alarm 6073 (dividing device not ready) will be released.

Remedy: lock dividing device

7270: OFFSET COMPENSATION ACTIVE !

Only with PC-MILL 105

Offset compensation activated by the following operation sequence.

- Reference point not active
- Machine in reference mode
- Key switch in manual operation
- Press STRG (or CTRL) and simultaneously 4

This must be carried out if prior to the tool change procedure spindle positioning is not completed (tolerance window too large)

7271: COMPENSATION FINISHED, DATA SAVED !

see 7270

PC TURN 105/120/125/155

The following alarms are valid for the lathes
PC TURN 105/120/125/155.

6000: EMERGENCY OFF

The EMERGENCY OFF key was pressed.
The reference position will be lost, the auxiliary drives will be switched off.
Remove the endangering situation and restart machine and software.

6001: PLC-CYCLE TIME EXCEEDING

The auxiliary drives will be switched off.
Contact EMCO Service.

6002: PLC - NO PROGRAM CHARGED

The auxiliary drives will be switched off.
Contact EMCO Service.

6003: PLC - NO DATA UNIT

The auxiliary drives will be switched off.
Contact EMCO Service.

6004: PLC - RAM MEMORY FAILURE

The auxiliary drives will be switched off.
Contact EMCO Service.

6008: MISSING CAN SUBSCRIBER

Check fuses or contact EMCO Service.

6009: SAFETY CIRCUIT FAULT

Defective step motor system.
A running CNC program will be interrupted, the auxiliary drives will be stopped, the reference position will be lost.
Contact EMCO Service.

6010: DRIVE X-AXIS NOT READY

The step motor board is defective or too hot, a fuse is defective, over- or undervoltage from mains.
A running program will be stopped, the auxiliary drives will be switched off, the reference position will be lost.
Check fuses or contact EMCO service.

6012: DRIVE Z-AXIS NOT READY

see 6010.

6013: MAIN DRIVE NOT READY

Main drive power supply defective or main drive too hot, fuse defective, over- or undervoltage from mains.
A running program will be stopped, the auxiliary drives will be switched off.
Check fuses or contact EMCO Service.

6014: NO MAIN SPINDLE SPEED

This alarm will be released, when the spindle speed is lower than 20 rpm because of overload.
Alter cutting data (feed, infeed, spindle speed).
The CNC program will be aborted, the auxiliary drives will be switched off.

6015: NO DRIVEN TOOL SPINDLE SPEED

see 6014.

6024: MACHINE DOOR OPEN

The door was opened while a machine movement.
The program will be aborted.

6040: TOOL TURRET INDEX FAILURE

The tool turret is in no locked position, tool turret sensor board defective, cabling defective, fuse defective.
A running CNC program will be stopped.
Swivel the tool turret with the tool turret key, check fuses or contact EMCO service.

6041: TOOL CHANGE TIMEOUT

Tool drum stuck (collision?), fuse defective, hardware defective.
A running CNC program will be stopped.
Check for collisions, check fuses or contact EMCO service.

6042: TOOL TURRET OVERHEAT

Tool turret motor too hot.

With the tool turret a max. of 14 swivel procedures a minute may be carried out.

6043: TOOL CHANGE TIMEOUT

Tool drum sticks (collision?), fuse defective, hardware defective.

A running CNC program will be stopped.

Check for collisions, check fuses or contact EMCO service.

6046: TOOL TURRET ENCODER FAULT

Fuse defective, hardware defective.

Check fuses or contact EMCO service.

6048: CHUCK NOT READY

Attempt to start the spindle with open chuck or without clamped workpiece.

Chuck sticks mechanically, insufficient pressured air supply, fuse defective, hardware defective.

Check fuses or contact EMCO service.

6049: COLLET NOT READY

see 6048

6050: M25 DURING SPINDLE ROTATION

With M25 the main spindle must stand still (consider run-out time, evtl. program a dwell)

6055: NO PART CLAMPED

This alarm occurs when with rotating spindle the clamping device or the tailstock reach the end position.

The workpiece has been pushed out of the chuck or has been pushed into the chuck by the tailstock.

Check clamping device settings, clamping forces, alter cutting data.

6056: QUILL NOT READY

Attempt to start the spindle or to move an axis or to swivel the tool turret with undefined tailstock position.

Tailstock is locked mechanically (collision), insufficient pressured air supply, fuse defective, magnetic switch defective.

Check for collisions, check fuses or contact EMCO service.

6057: M20/M21 DURING SPINDLE ROTATION

With M20/M21 the main spindle must stand still (consider run-out time, evtl. program a dwell)

6058: M25/M26 DURING QUILL FORWARD

To actuate the clamping device in an NC program with M25 or M26 the tailstock must be in back end position.

6059: C-AXIS SWING IN TIMEOUT

C-axis does not swivel in within 4 seconds.

Reason: not sufficient air pressure, and/or mechanics stuck.

6060: C-AXIS INDEX FAILURE

When swivelling in the C-axis the limit switch does not respond.

Check pneumatics, mechanics and limit switch.

6064: AUTOMATIC DOOR NOT READY

Door sticks mechanically (collision), insufficient pressured air supply, limit switch defective, fuse defective.

Check for collisions, check fuses or contact EMCO service.

6065: LOADER MAGAZINE FAILURE

Loader not ready.

Check if the loader is switched on, correctly connected and ready for operation and/or disable loader (WinConfig).

6066: CLAMPING DEVICE FAILURE

No compressed air at the clamping device

Check pneumatics and position of the clamping device proximity detectors.

7000: INVALID TOOL NUMBER PROGRAMMED

The tool position was programmed larger than 8.

The CNC program will be stopped.

Interrupt program with RESET and correct the program.

7016: SWITCH ON AUXILIARY DRIVES

The auxiliary drives are off. Press the AUX ON key for at least 0.5 sec. (to avoid accidentally switching on) to switch on the auxiliary drives (also a lubricating pulse will be released).

7017: REFERENCE MACHINE

Approach the reference point.

When the reference point is not active, manual movements are possible only with key switch at position "setting operation".

7018: TURN KEY SWITCH

With NC-Start the key switch was in position "setting operation".

NC-Start is locked.

Turn the key switch in the position "automatic" to run a program.

7019: PNEUMATIC LUBRICATION MONITORING!

Refill pneumatic oil

7020: SPECIAL OPERATION MODE ACTIVE

Special operation mode: The machine door is opened, the auxiliary drives are switched on, the key switch is in position "setting operation" and the consent key is pressed.

Manual traversing the axes is possible with open door. Swivelling the tool turret is possible with open door. Running a CNC program is possible only with standing spindle (DRYRUN) and SINGLE block operation.

For safety: If the consent key is pressed for more than 40 sec. the function of this key is interrupted, the consent key must be released and pressed again.

7021: TOOL TURRET NOT LOCKED

The tool turret operating was interrupted.

NC start and spindle start are locked. Press the tool turret key in the RESET status of the control.

7038: LUBRICATION SYSTEM FAULT

The pressure switch is defective or gagged.

NC-Start is locked. This alarm can be reset only by switching off and on the machine.

Contact EMCO service.

7039: LUBRICATION SYSTEM FAULT

Not enough lubricant, the pressure switch is defective.

NC-Start is locked.

Check the lubricant and lubricate manually or contact EMCO service.

7040: MACHINE DOOR OPEN

The main drive can not be switched on and NC-Start can not be activated (except special operation mode) Close the machine to run a program.

7042: INITIALIZE MACHINE DOOR

Every movement and NC-Start are locked.

Open and close the machine door to initialize the safety circuits.

7043: PIECE COUNT REACHED

A predetermined number of program runs was reached. NC-Start is locked. Reset the counter to continue.

7048: CHUCK OPEN

This message shows that the chuck is open. It will disappear if a workpiece will be clamped.

7049: CHUCK - NO PART CLAMPED

No part is clamped, the spindle can not be switched on.

7050: COLLET OPEN

This message shows that the collet is open. It will disappear if a workpiece will be clamped.

7051: COLLET - NO PART CLAMPED

No part is clamped, the spindle can not be switched on.

7052: QUILL IN UNDEFINED POSITION

The tailstock is in no defined position.

All axis movements, the spindle and the tool turret are locked.

Drive the tailstock in back end position or clamp a workpiece with the tailstock.

7053: QUILL - NO PART CLAMPED

The tailstock reached the front end position. Traverse the tailstock back to the back end position to continue.

7054: NO PART CLAMPED

No part clamped, switch-on of the spindle is locked.

7055: CLAMPING DEVICE OPEN

This message indicates that the clamping device is not in clamping state. It disappears as soon as a part is clamped.

AC95 ALARMS

Axis Controller Alarms 8000 - 9999

8000 Fatal Error AC

8004 ORDxx Failure main-drive unit

8005 - 8009 ORDxx Internal error AC

Remedy: report to EMCO if repeatable

8010 ORDxx Syncr. error main drive

Cause: synchronisation mark missing for main drive

Remedy: report to service technician, if repeatable

8011 - 8013 ORDxx Internal error AC

Remedy: report to EMCO, if repeatable

8014 ORDxx Decel.-time of axis too high

Remedy: report to service technician, if repeatable

8018 ORDxx Internal error AC

Remedy: report to EMCO, if repeatable

8021 ORDxx Internal error AC

Remedy: report to EMCO, if repeatable

8022 ORDxx Internal error AC

Remedy: report to EMCO, if repeatable

8023 ORDxx Invalid Z value for helix

Cause: The Z value of the helix must be smaller than the length of the arc to be traversed

Remedy: Program correction

8100 Fatal init error AC

Cause: Internal error

Remedy: Restart software or reinstall when necessary, report to EMCO, if repeatable.

8101 Fatal init error AC

see 8101.

8102 Fatal init error AC

see 8101.

8103 Fatal init error AC

see 8101.

8104 Fatal system error AC

see 8101.

8105 Fatal init error AC

see 8101.

8106 No PC-COM card found

Cause: PC-COM board can not be accessed (ev. not mounted).

Remedy: Mount board, adjust other address with jumper

8107 PC-COM card not working

see 8106.

8108 Fatal error on PC-COM card

see 8106.

8109 Fatal error on PC-COM card

see 8106.

8110 PC-COM init message missing

Cause: Internal error

Remedy: Restart software or reinstall when necessary, report to EMCO, if repeatable.

8111 Wrong configuration of PC-COM

see 8110.

8113 Invalid data (pccom.hex)

see 8110.

8114 Programming error on PC-COM

see 8110.

8115 PC-COM packet acknowledge missing

see 8110.

8116 PC-COM startup error

see 8110.

8117 Fatal init data error (pccom.hex)

see 8110.

8118 Fatal init error AC

see 8110, ev. insufficient RAM memory

8119 PC interrupt no. not valid

Cause: The PC interrupt number can not be used.

Remedy: Find out free interrupt number in the Windows95 system control (allowed: 5,7,10, 11, 12, 3, 4 und 5) and enter this number in WinConfig.

8120 PC interrupt no. unmaskable

see 8119

8121 Invalid command to PC-COM

Cause: Internal error or defective cable

Remedy: Check cables (screw it); Restart software or reinstall when necessary, report to EMCO, if repeatable.

8122 Internal AC mailbox overrun

Cause: Internal error

Remedy: Restart software or reinstall when necessary, report to EMCO, if repeatable.

8123 Open error on record file

Cause: Internal error

Remedy: Restart software or reinstall when necessary, report to EMCO, if repeatable.

8124 Write error on record file

Cause: Internal error

Remedy: Restart software or reinstall when necessary, report to EMCO, if repeatable.

8125 Invalid memory for record buffer

Cause: Insufficient RAM, record time exceeding.
 Remedy: Restart software, ev. remove drivers etc. to gain more RAM, reduce record time.

8126 AC Interpolation overrun

Cause: Ev. insufficient computer performance.
 Remedy: Set a longer interrupt time in WinConfig. This may result in poorer path accuracy.

8127 Insufficient memory

Cause: Insufficient RAM
 Remedy: Close other programs, restart software, ev. remove drivers etc. to gain more RAM.

8128 Invalid message to AC

Cause: Internal error
 Remedy: Restart software or reinstall when necessary, report to EMCO, if repeatable.

8129 Invalid MSD data - axisconfig.

see 8128.

8130 Internal init error AC

see 8128.

8130 Internal init error AC

see 8128.

8132 Axis accessed by multiple channels

see 8128.

8133 Insufficient NC block memory AC

see 8128.

8134 Too much center points programmed

see 8128.

8135 No centerpoint programmed

see 8128.

8136 Circle radius too small

see 8128.

8137 Invalid for Helix specified

Cause: Wrong axis for helix. The combination of linear and circular axes does not match.
 Remedy: Program correction.

8140 Maschine (ACIF) not responding

Cause: Machine off or not connected.
 Remedy: Switch on machine or connect.

8141 Internal PC-COM error

Cause: Internal error
 Remedy: Restart software or reinstall when necessary, report to EMCO, if repeatable.

8142 ACIF Program error

Cause: Internal error
 Remedy: Restart software or reinstall when necessary, report to EMCO, if repeatable.

8143 ACIF packet acknowledge missing

see 8142.

8144 ACIF startup error

see 8142.

8145 Fatal init data error (acif.hex)

see 8142.

8146 Multiple request for axis

see 8142.

8147 Invalid PC-COM state (DPRAM)

see 8142.

8148 Invalid PC-COM command (CNo)

see 8142.

8149 Invalid PC-COM command (Len)

see 8142.

8150 Fatal ACIF error

see 8142.

8151 AC Init Error (missing RPG file)

see 8142.

8152 AC Init Error (RPG file format)

see 8142.

8153 FPGA program timeout on ACIF

see 8142.

8154 Invalid Command to PC-COM

see 8142.

8155 Invalid FPGA packet acknowledge

see 8142 or hardware error on ACIF board (contact EMCO Service).

8156 Sync within 1.5 revol. not found

see 8142 or Bero hardware error (contact EMCO Service).

8157 Data record done

see 8142.

8158 Bero width too large (referencing)

see 8142 or Bero hardware error (contact EMCO Service).

8159 Function not implemented

Bedeutung: In normal operation this function can not be executed

8160 Axis synchronization lost axis 3..7

Cause: Axis spins or slide is locked, axis synchronisation was lost

Remedy: Approach reference point

8161 X-Axis synchronization lost

Step loss of the step motor. Causes:

- Axis mechanically blocked
- Axis belt defective
- Distance of proximity detector too large (>0,3mm) or proximity detector defective
- Step motor defective

8162 Y-Axis synchronization lost

see 8161

8163 Z-Axis synchronization lost

see 8161

8164 Software limit switch max axis 3..7

Cause: Axis is at traverse area end

Remedy: Retract axis

8168 Software limit overtravel axis 3..7

Cause: Axis is at traverse area end

Remedy: Retract axis

8172 Communication error to machine

Cause: Internal error

Remedy: Restart software or reinstall when necessary, report to EMCO, if repeatable.
Check connection PC - machine, eventually eliminate distortion sources.

8173 INC while NC program is running**8174 INC not allowed****8175 MSD file could not be opened**

Cause: Internal error

Remedy: Restart software oder bei Bedarf neu installieren, report to EMCO, if repeatable.

8176 PLS file could not be opened

see 8175.

8177 PLS file could not be accessed

see 8175.

8178 PLS file could not be written

see 8175.

8179 ACS file could not be opened

see 8175.

8180 ACS file could not be accessed

see 8175.

8181 ACS file could not be written

see 8175.

8182 Gear change not allowed**8183 Gear too high****8184 Invalid interpolaton command****8185 Forbidden MSD data change**

see 8175.

8186 MSD file could not be opened

see 8175.

8187 PLC program error

see 8175.

8188 Gear command invalid

see 8175.

8189 Invalid channel assignement

see 8175.

8190 Invalid channel within message**8191 Invalid jog feed unit****8192 Invalid axis in command****8193 Fatal PLC error**

see 8175.

8194 Thread without length**8195 No thread slope in leading axis**

Remedy: Program thread pitch

8196 Too many axis for thread

Remedy: Program max. 2 axes for thread.

8197 Thread not long enough

Cause: Thread length too short.

With transition from one thread to the other the length of the second thread must be sufficient to produce a correct thread.

Remedy: Longer second thread or replace it by a linear interpolation (G1).

8198 Internal error (to many threads)

see 8175.

8199 Internal error (thread state)

Cause: Internal error

Remedy: Restart software or reinstall when necessary, report to EMCO, if repeatable.

8200 Thread without spindle on

Remedy: Switch on spindle

8201 Internal thread error (IPO)

see 8199.

8201 Internal thread error (IPO)

see 8199.

8203 Fatal AC error (0-ptr IPO)

see 8199.

8204 Fatal init error: PLC/IPO running

see 8199.

8205 PLC Runtime exceeded

Cause: Insufficient computer performance

8206 Invalid PLC M-group initialisation

see 8199.

8207 Invalid PLC machine data

see 8199.

8208 Invalid application message

see 8199.

8211 Feed too high (thread)

Cause: Thread pitch too large / missing, Feed for thread reaches 80% of rapid feed

Remedy: Program correction, lower pitch or lower spindle speed for thread

8212 Rotation axis not allowed**8213 Circle and rotation axis can't be interpolated****8214 Thread and rotation axis can't be interpolated****8215 Invalid state****8216 No rotation axis for rotation axis switch****8217 Axis type not valid!****8218 Referencing round axis without selected round axis!****8219 Thread not allowed without spindle encoder!****8220 Buffer length exceeded in PC send message!****8221 Spindle release although axis is no spindle!****8222 New master spindle is not valid****8223 Can't change master spindle (no M5)!****8224 Invalid stop mode****8225 Invalid parameter for BC_MOVE_TO_IO!****8226 Rotary axis switch not valid (MSD data)!****8227 Speed setting not allowed while rotary axis is active!****8228 Rotary axis switch not allowed while axis move!****8229 Spindle on not allowed while rotary axis is active!****8230 Program start not allowed due to active spindle rotation axis!****8231 Axis configuration (MSD) for TRANSMIT not valid!****8232 Axis configuration (MSD) for TRACYL not valid!****8233 Axis not available while TRANSMIT/TRACYL is active!****8234 Axis control grant removed by PLC while axis interpolates!****8235 Interpolation invalid while axis control grant is off by PLC!****8236 TRANSMIT/TRACYL activated while axis or spindle moves!****8237 Motion through pole in TRANSMIT!****8238 Speed limit in TRANSMIT exceeded!****8239 DAU exceeded 10V limit!****8240 Function not valid during active transformation (TRANSMIT/TRACYL)!****8241 TRANSMIT not enabled (MSD)!****8242 TRACYL not enabled (MSD)!****8243 Round axis invalid during active transformation!****8245 TRACYL radius = 0!****8246 Offset alignment not valid for this state!****8247 Offset alignment: MSD file write protected!****8248 Cyclic supervision failed!****8249 Axis motion check alarm!****8250 Spindle must be rotation axis !****8251 Lead for G331/G332 missing !****8252 Multiple or no linear axis programmed for G331/G332 !****8253 Speed value for G331/G332 and G96 missing !****8254 Value for thread starting point offset not valid!****8255 Reference point not in valid software limits!****8256 Spindle speed too low while executing G331/G332!**

I: Control Alarms

Control Alarms

These alarm s can occur only with operating and programming the control functions or with running CNC programs.

1 RS232 parity error !

Cause: Data transmission error parity error, wrong RS 232 setting in external device

Remedy: Check data cables, set serial interface of the external device

2 RS232 transmission error !

Cause: Data transmission error character overflow
Data transmission error invalid data frame

Remedy: Check data cables, set serial interface of the external device

10 Nxxxx Invalid G-code

Remedy: Program correction

11 ORDxx Feed wrong/missing

Cause: Attempt to start with feed = 0, also with G95/96, if S = 0 or M5

Remedy: Program correction

21 Nxxxx Circle: Wrong plane selected

Cause: The wrong plane (G17, 18, 19) is active for a circle

Remedy: Program correction

30 Nxxxx Invalid tool offset number

Cause: The lower 2 digits of the T number are to great

Remedy: Program correction

33 Nxxxx CRC can't be determined

Cause: Too much blocks without new position programmed, invalid contour element, programmed circle radius smaller than cutter radius, contour element to short.

Remedy: Program correction

34 Nxxxx Error on deactivating CRC

Remedy: Program correction

37 Nxxxx Plane change while CRC act.

Cause: Change of plane not permitted with active cutter radius compensation

Remedy: Program correction

41 Nxxxx Contour violation CRC

Cause: Invalid contour element, programmed circle radius smaller than cutter radius, contour element to short, contour violation with full circle.

Remedy: Program correction

51 Nxxxx Wrong chamfer/radius value

Cause: The contour elements between a chamfer / radius should be inserted are too short.

Remedy: Program correction

52 Nxxxx Invalid contour draft

Cause: From the programmed parameters no valid contour draft would result

Remedy: Program correction

53 Nxxxx Wrong parameter structure

Cause: From the programmed parameters no valid contour draft would result, wrong parameter programmed

Remedy: Program correction

56 Nxxxx Wrong angle value

Cause: With the programmed angle no intersection point would result

Remedy: Program correction

57 Nxxxx Error in contour draft

Cause: Invalid parameters programmed.

Remedy: Program correction

58 Nxxxx Contour draft not determinable

Cause: Too much blocks without new position programmed, program end while contour draft

Remedy: Program correction

60 Nxxxx Block number not found

Cause: Jump target not found

Remedy: Program correction

62 Nxxxx General cycle error

Cause: Call-up counter of subprogram call invalid, feed<=0, thread pitch missing/<=0, cutting depth missing/<=0/invalid, retraction height to small, block address P/Q missing, declaration pattern repetition missing/invalid, infeed for next cut missing/invalid, undercut at cycle ground <0, cycle end point missing/invalid, thread end point missing/invalid;

Remedy: Program correction

63 Nxxxx Wrong Cycle call

Cause: P/Q missing, wrong address

Remedy: Program correction

70 Insufficient memory

Cause: The PC has not enough memory

Remedy: Close all other Windows applications, remove resident programs from memory, restart the PC

71 Program not found

Cause: NC program not found
With program start no program was selected
Remedy: Correct call-up or create program, select program

73 File already exists !

Remedy: Select other file name.

77 Insufficient RAM for subroutine

Cause: Subprograms interlocked too deep
Remedy: Program correction

83 Nxxxx Circle not in active plane

Cause: Circle is not in active plane for CRC
Remedy: Program correction

142 Wrong simulation area

Cause: Wrong scale factor (e.g. 0) programmed
Remedy: Program correction

142 Invalid scale factor

Cause: No or an invalid simulation area was entered
Remedy: Enter correct simulation area

315 ORDxx Rotatory checking X

Cause: The step motor has fallen out of pace
Remedy: Reduce infeed and feed, check slides for smooth running, approach reference point

325 ORDxx Rotatory checking Y

see alarm 315

335 ORDxx Rotatory checking Z

see alarm 315

500 ORDxx Target point exceeds work.area

Cause: Target point, circle target point or circle out of working area limitation
Remedy: Program correction

501 ORDxx Target point exceeds SW limit

Cause: Target point, circle target point or circle out of working area limitation
Remedy: Program correction

510 ORDxx Software-limit switch X

Cause: Software limit switch in X exceeded (JOG)
Remedy: Traverse back manually

520 ORDxx Software-limit switch Y

see 510

530 ORDxx Software-limit switch Z

see 510

2501 ORDxx Synchronisation-error AC

Remedy: RESET, report to EMCO if reproducible

2502 ORDxx Synchronisation-error AC

see 2501

2503 ORDxx Synchronisation-error AC

see 2501

2504 ORDxx No memory for interpreter

Cause: Too less RAM memory, continueing the program is not possible
Remedy: Close all Windows application, close WinNC, remove resident programs from AUTOEXEC.BAT and CONFIG.SYS, restart the PC

2505 ORDxx No memory for interpreter

see 2504

2506 ORDxx Too less RAM

see 2504

2507 ORDxx Reference point not active

Remedy: Approach reference point

2508 ORDxx Internal error NC core

Remedy: RESET, report to EMCO if reproducible

2520 ORDxx RS485 device absent

Cause: With program start a RS485 device did not report, while program run a device got defective
AC Axis controller
SPS PLC
MT control keyboard
Remedy: Switch on RS485 device (machine, control keyboard), check cables and plugs, check terminator plug, report to EMCO if reproducible

2521 ORDxx RS485 communication error

Remedy: PC restart, report to EMCO if reproducible

2522 ORDxx RS485 communication error

Remedy: PC restart, report to EMCO if reproducible

2523 ORDxx INIT error on RS485 PC-board

See "Software Installation", Mistakes with installation of the software

2524 ORDxx Gen.-Failure RS485 PC-board

Remedy: PC restart, report to EMCO if reproducible

2525 ORDxx Transmit error RS485

Cause: Transmission error by poor plug connections, missing terminator, external sources of electromagnetic interference
Remedy: Check the error sources above

2526 ORDxx Transmit error RS485

see 2525

2527 ORDxx Internal error AC

Remedy: Switch machine off/on, report to EMCO if reproducible

2528 ORDxx Operating system error PLC

Remedy: Switch machine off/on, report to EMCO if reproducible

2529 ORDxx External keyboard error

Remedy: The external keyboard always must be switched on after the PC. Restart the software, report to EMCO if reproducible

2540 ORDxx Error saving setting-data

Cause: Hard disk full, wrong path setting, no writing access

Remedy: Check hard disk space, check writing access, reinstallation of the software if reproducible

2545 ORDxx Drive / Device not ready

Remedy: Insert disk, lock drive, check disk drive, ...

2546 ORDxx Checksum error machine-data

Remedy: Restart, report to EMCO if reproducible

2550 ORDxx PLC simulation error

Remedy: Restart, report to EMCO if reproducible

2551 ORDxx PLC simulation error

Remedy: Restart, report to EMCO if reproducible

2562 Read error on CNC program

Cause: Defective program file, DOS read error (disk, hard disk)

Remedy: Solve problem on DOS level, eventually reinstallation of the software

2614 ORDxx Internal error MSD

Remedy: Report to EMCO if reproducible

2650 ORDxx Internal error cycle call up

Cause: Invalid cycle call when a cycle was called with a G command

Remedy: Program correction

2849 Internal error CRC

Remedy: Report to EMCO if reproducible

2904 Helix Z value too large

Cause: The pitch of the helix must not be larger than 45°

Remedy: Program correction

